

Sewelo stone, found in Botswana 2019: 1756 ct (350g)
Source: Financial Times

EART 26001
Energy and Resources

Dr Stefan Schroeder

Lecture 5-1: Diamonds



2nd largest gem-quality diamond ever(?), found at Karowe mine, Botswana 2024: 2492 ct (498g)
Source: Guardian

Energy and Resources
Learning goals week 5

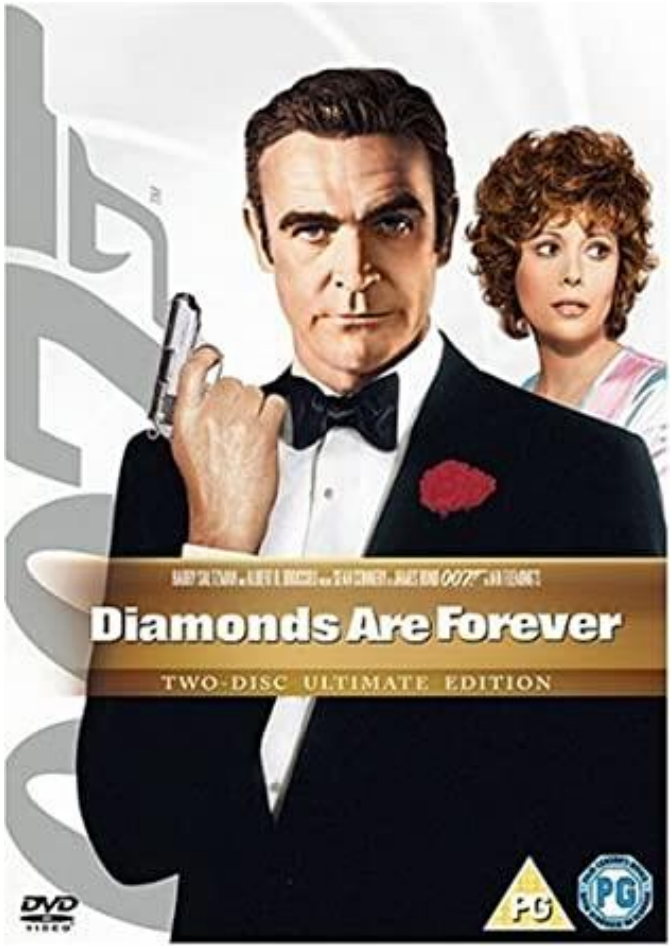
This week you will learn to:

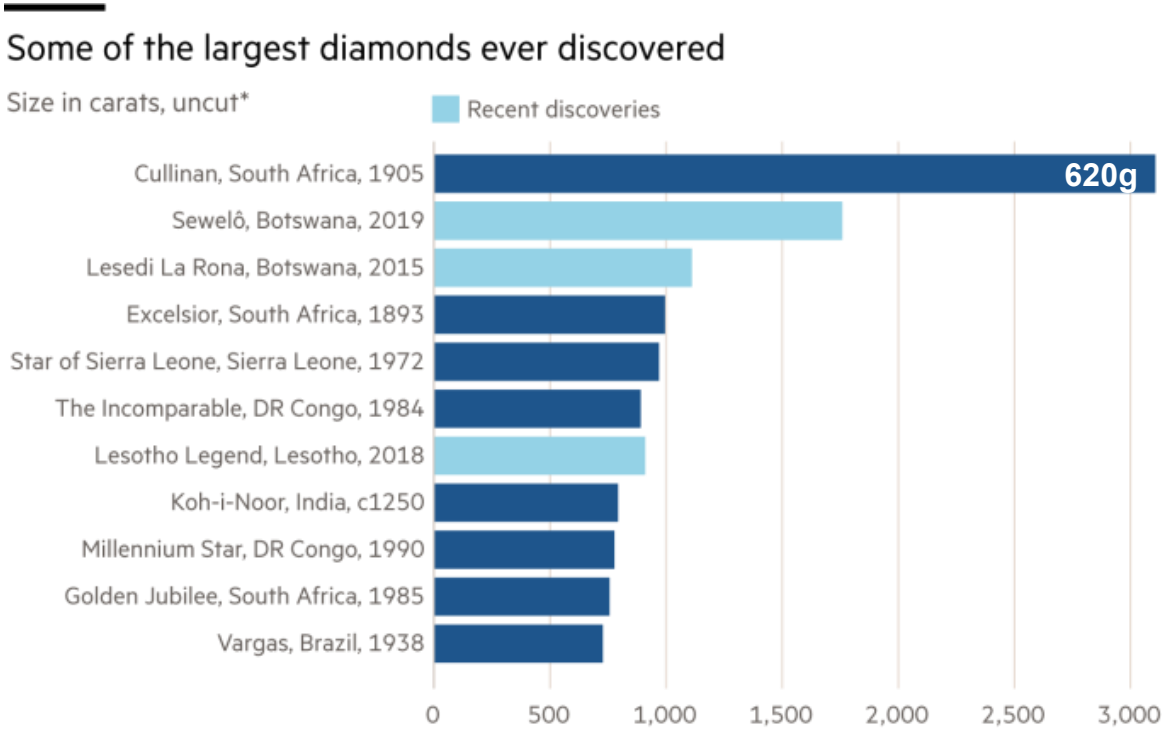
- **Describe the uses, physical and chemical properties of diamonds.**
- **Recognize the igneous and sedimentary processes forming diamond deposits.**
- **Explain the geotectonic context for diamond formation.**
- **Recognize the exploration and production methods of diamond mining.**

Diamond applications

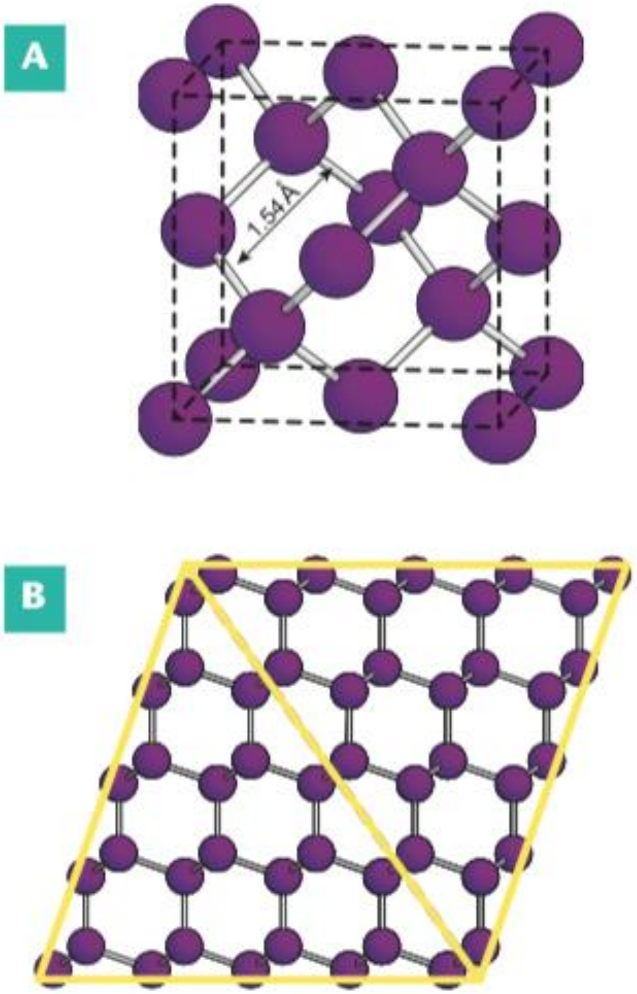
- Jewelry
- Industrial applications: cutting, grinding







Diamond properties

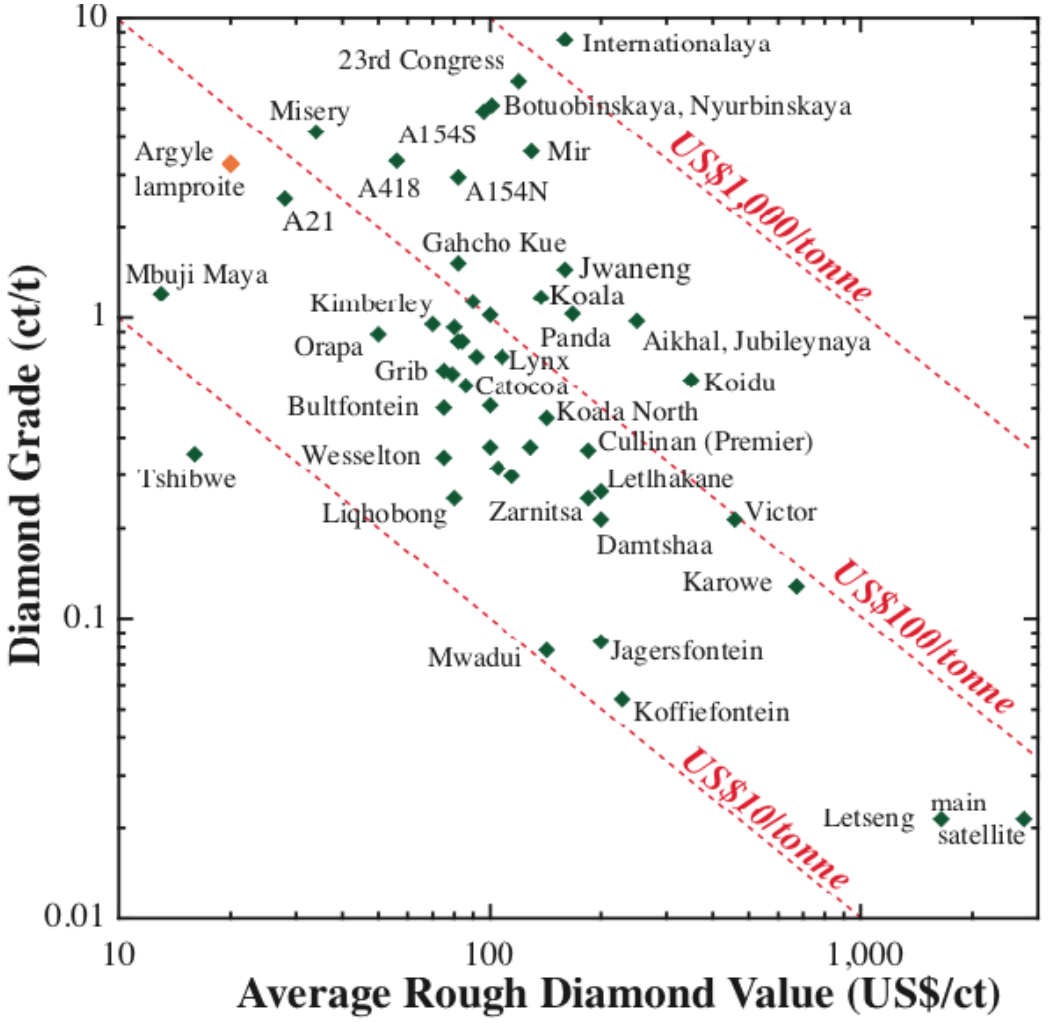


Composition	C (carbon)
Crystallographic class	Cubic – hexoctahedral (highest of symmetries)
Space group	$Fd3m$ $a = 3.57$ Å (cell edge)
Common crystal forms {and indices}	Octahedron {111}, cube {100}, dodecahedron {110}, rounded variations due to etching
Twins	Spinel-law common, yielding the flat triangular "macle"
Hardness	10 on Mohs' scale, 56–115 Knoop hardness number (GPa), 10,000 Brook's indenter scale, octahedral face hardest, cube face softest
Moduli	Bulk modulus: ~500 GPa; Young's modulus: ~1050 GPa
Cleavage	Excellent parallel to octahedron face {111}
Density	3.51 g cm^{-3} (or specific gravity = 3.51)
Luster	Adamantine (the definition for this kind of luster)
Colors	Colorless, yellow, blue, green, and many others
Refractive index	2.4175 (in the yellow light of a sodium lamp)
Dispersion	Large (0.0437 – the difference in index at G and B Fraunhofer wavelengths), leading to rainbow colors on refraction
Optical transmission	Transparent over a broad range of the electromagnetic spectrum; an excellent material for optical windows
Thermal conductivity	Superb, 5 to 25 watts centimeter^{-1} °C^{-1} (at 300K); 4 times greater than copper; an excellent thermal conductor
Electrical conductivity	0 to ~100 ohm cm^{-1} (resistivity at 300K); an insulator

Harlow 2005

Diamond
Mining grades

- Size of diamonds measured in carat – 1 ct = 200 mg
- Viability of diamond deposit depends on grade and value/quality of individual stones
- Grades typically 0.05-2 ct/t
  - Canada 3.9 ct/t, value $95/ct
  - Lesotho 0.01 ct/t, value $1000/ct
- Individual mines have different proportion of gem-quality stones
- alluvial diamonds have generally higher proportion of gem-quality stones

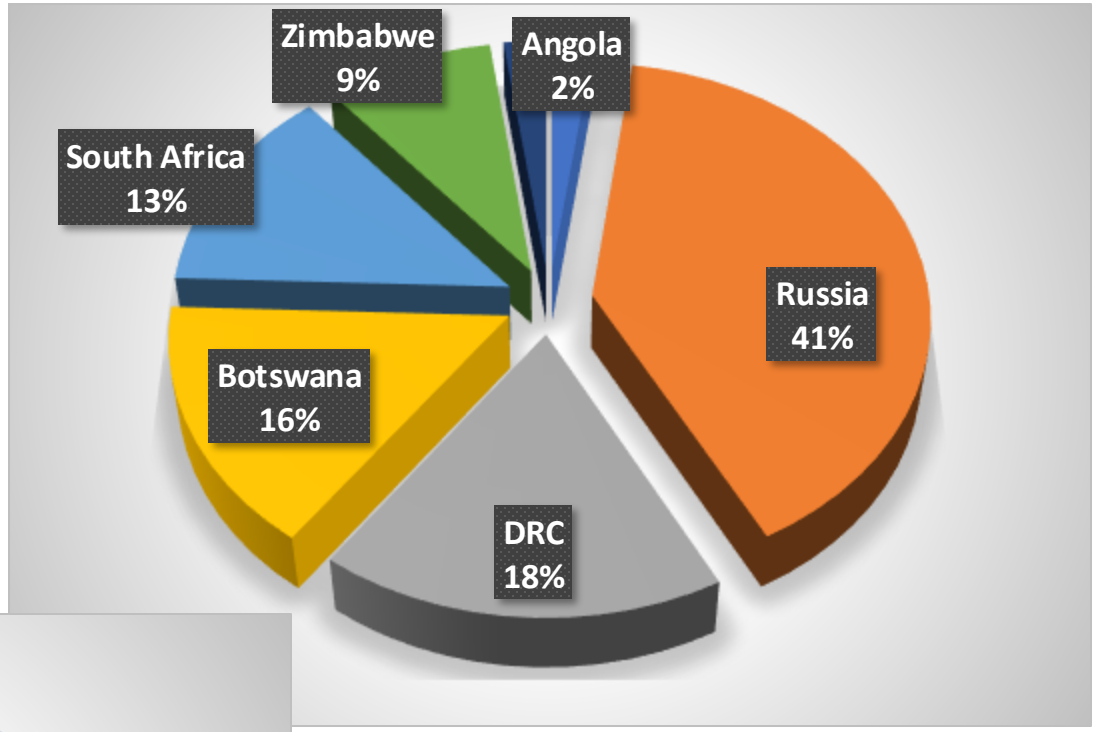


Kjarsgaard 2019

Diamond
Reserves

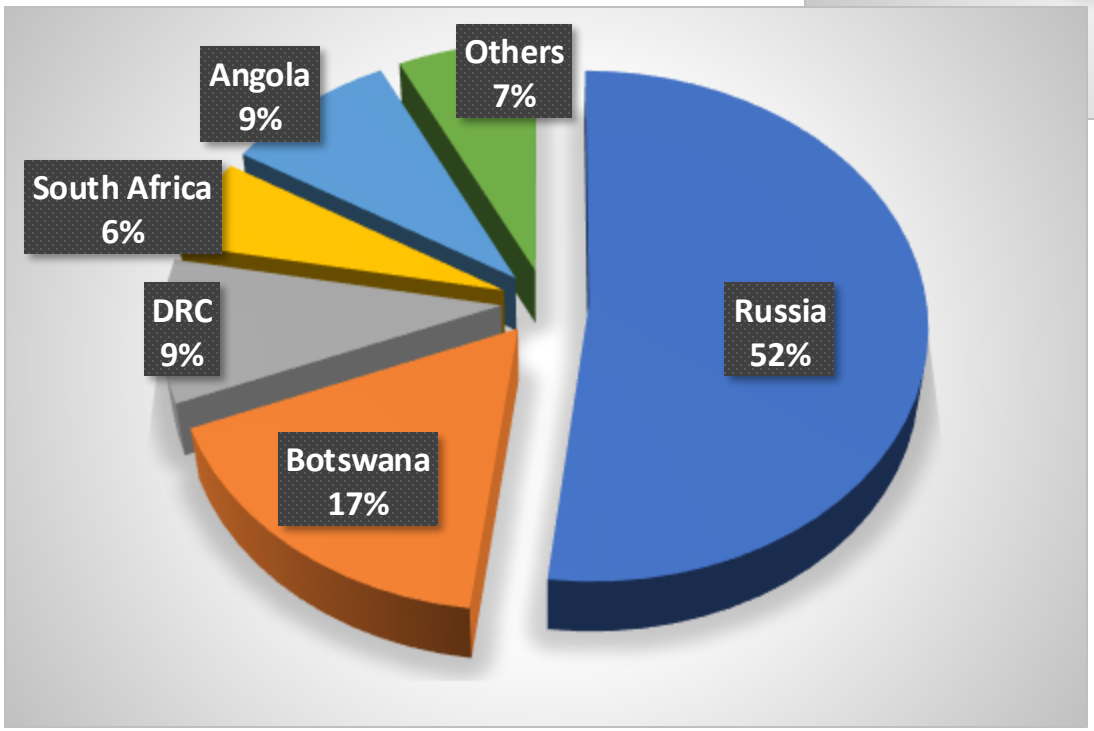
- Pie charts are by weight
- ‘ranking’ by value: Botswana, Russia, Angola, Canada
- China is main producer for synthetic diamonds

Natural industrial diamond mine production 2023



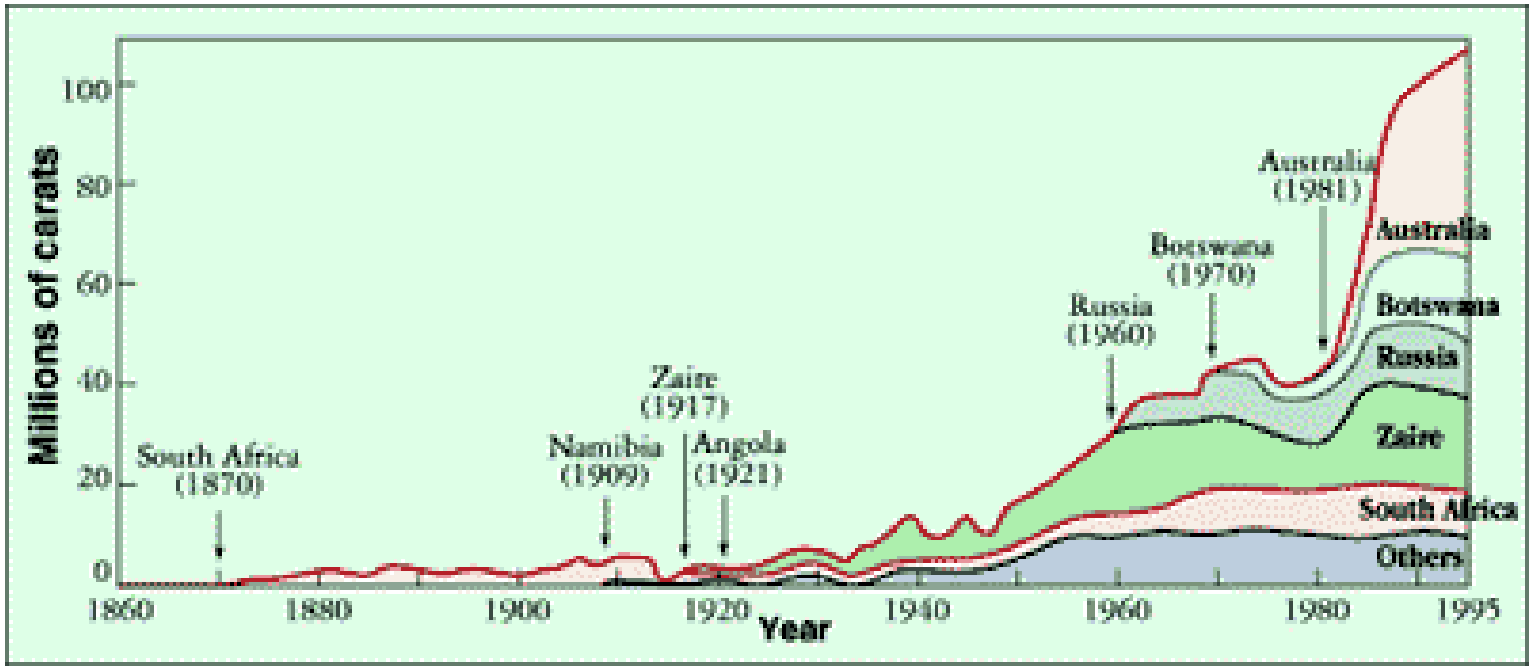
Source: USGS

Natural industrial diamond reserves 2023



Source: USGS

Diamond
Reserves



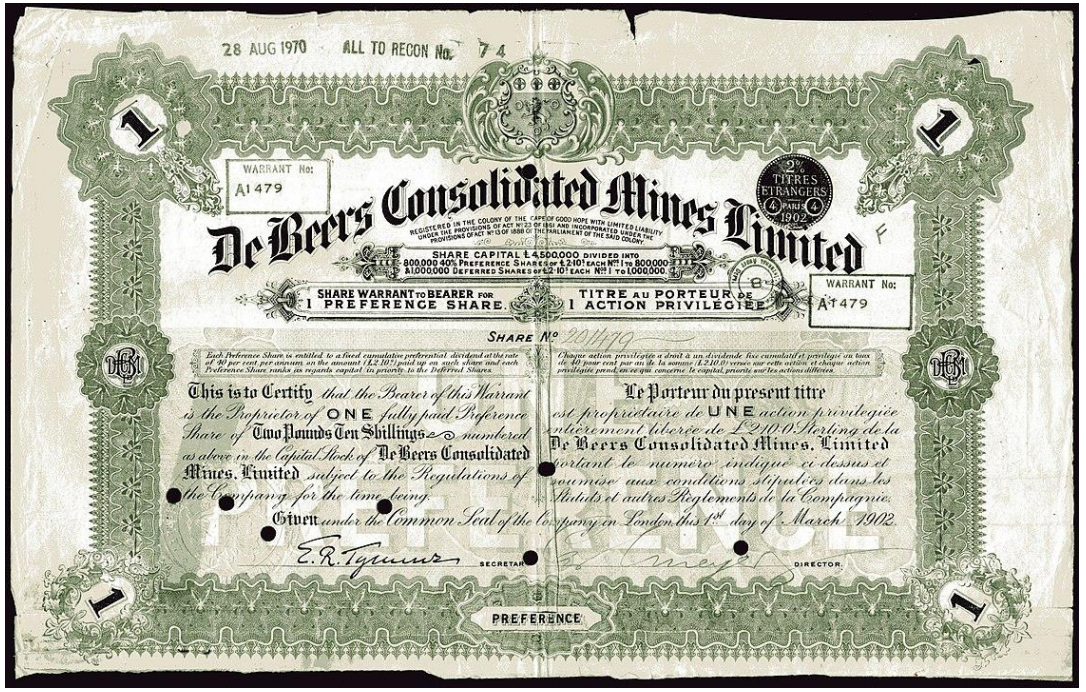
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Diamond
Market



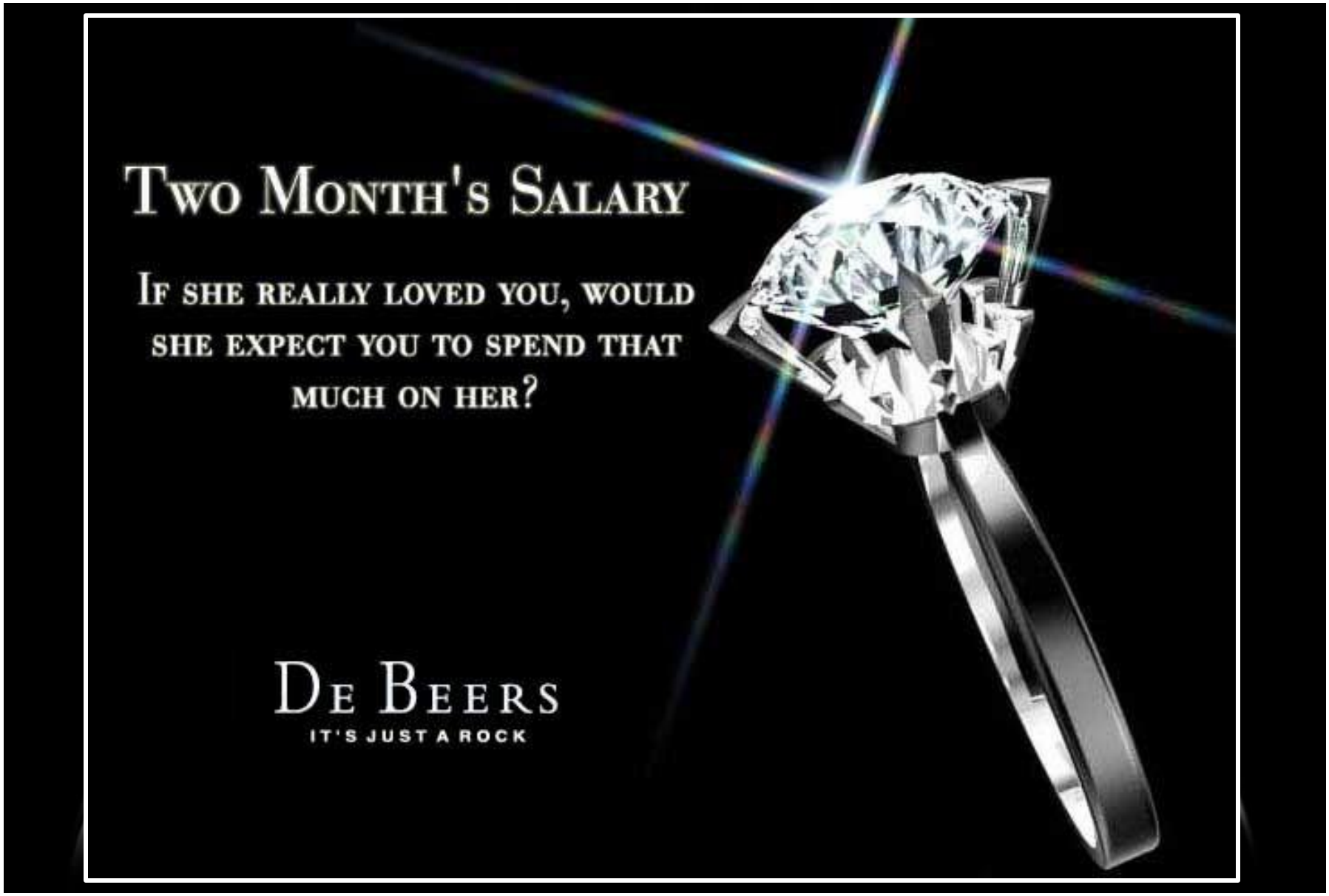


Premier Diamond Mine, South Africa. An illustration of extensive "open cut" mining



© mindat.org

Diamond
Market



Diamond
Market

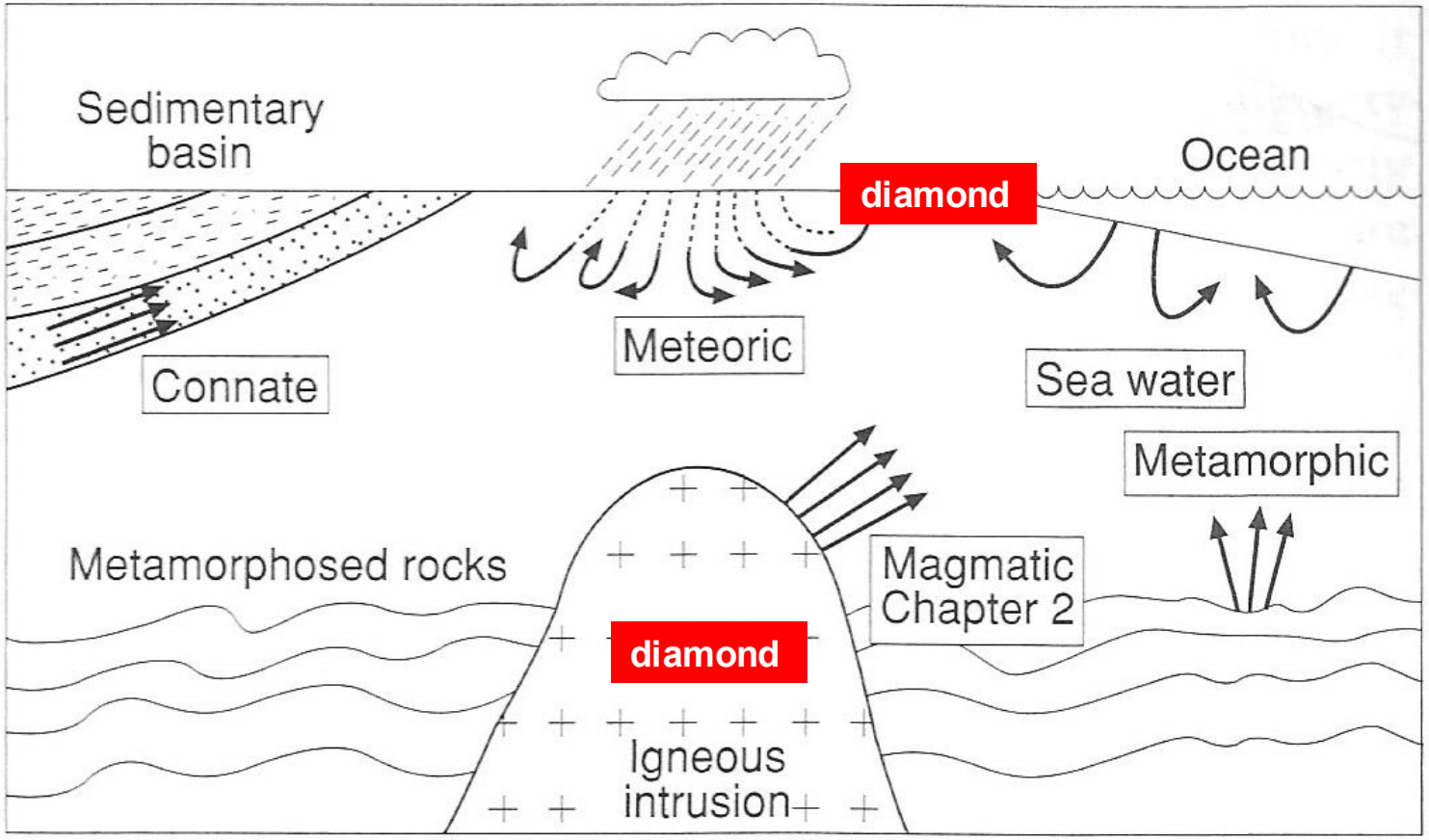




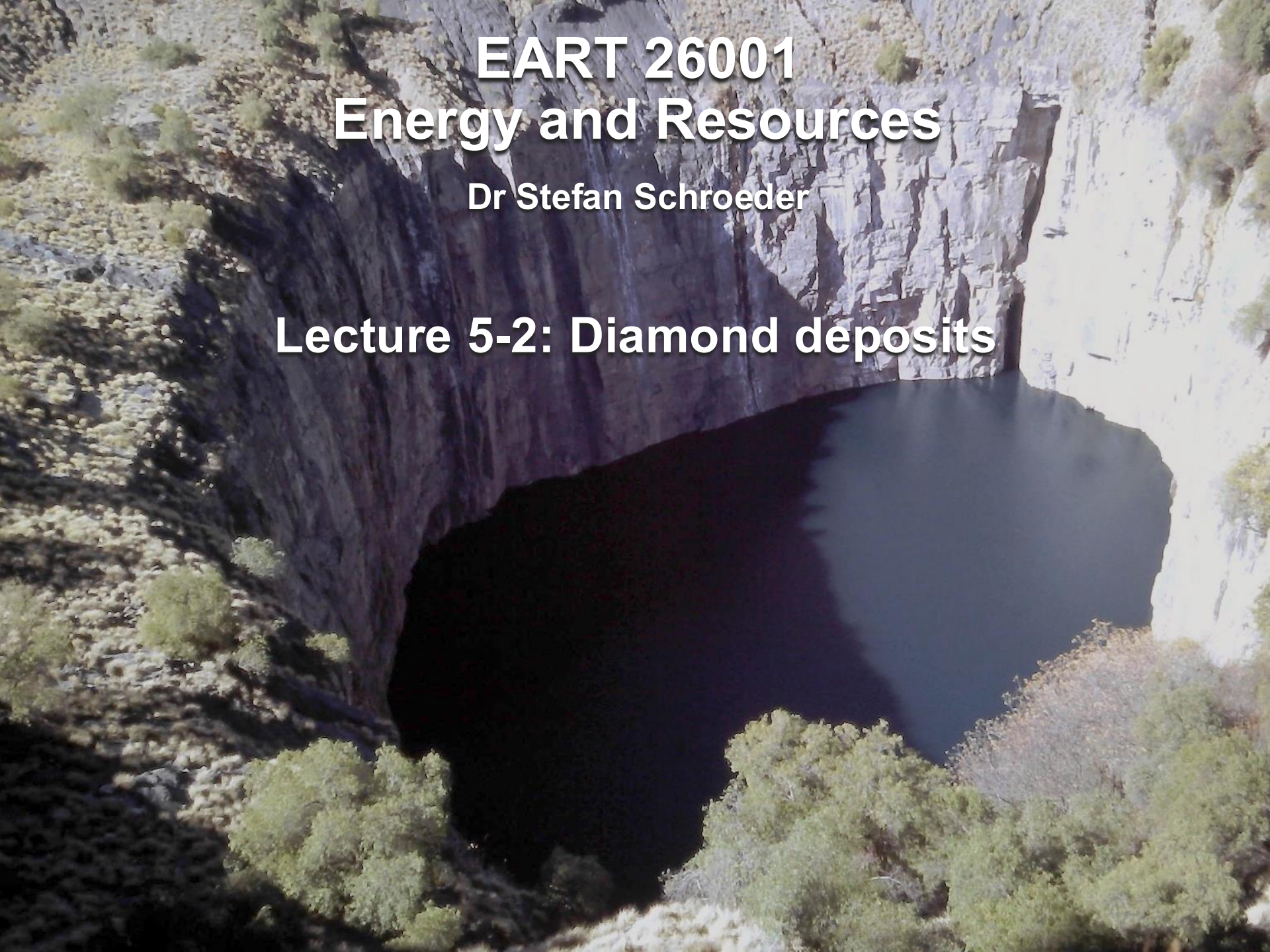


Diamond
Deposit types

- **Magmatic deposits**
  - **Volcanic pipes (kimberlites)**
- **Sedimentary deposits**
  - **Placers**



Robb 2008



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Lecture 5-2: Diamond deposits

Deposit types – kimberlite pipes
Lithology

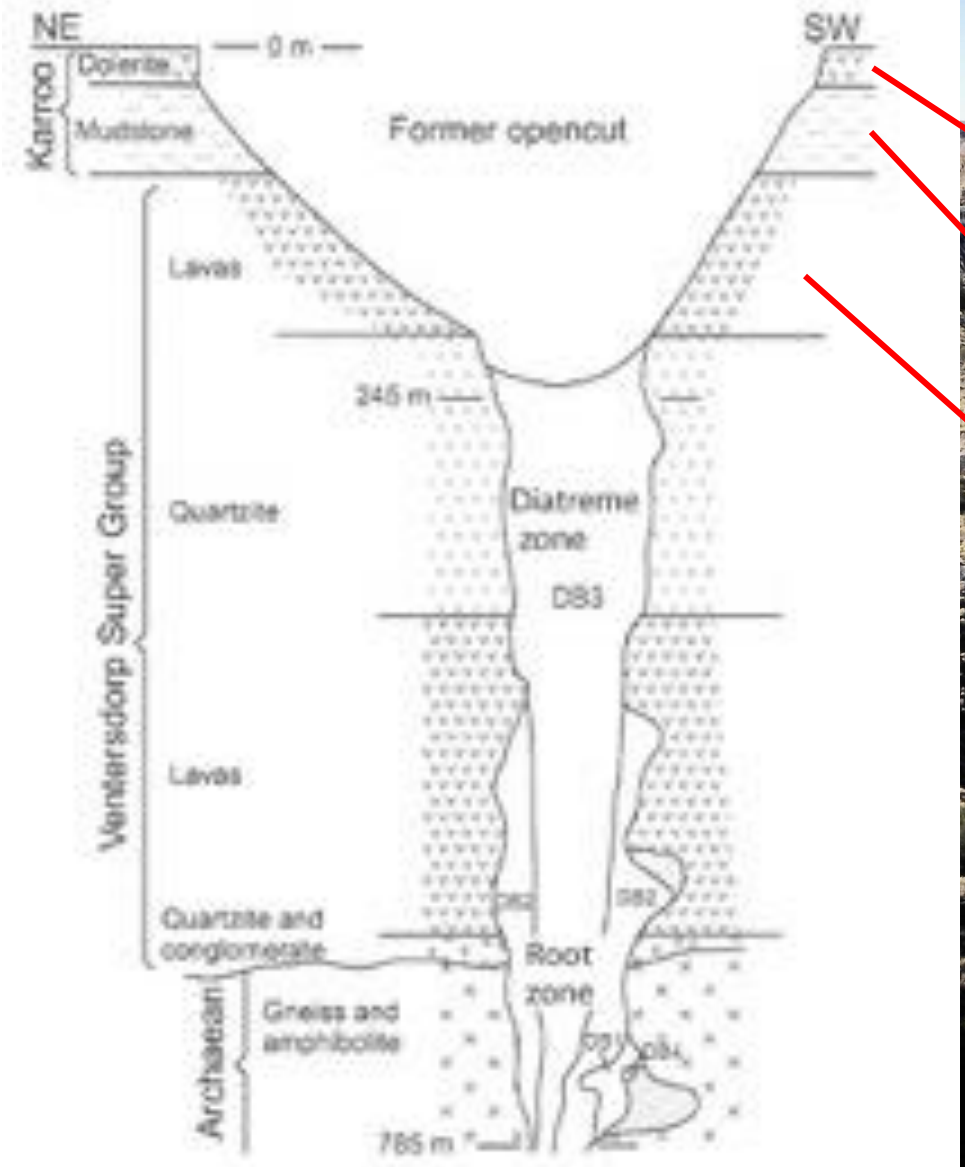




© www.streckeisen.it

- Kimberlite: altered, K- and Mg-rich ultramafic rock with phenocrysts of olivine, phlogopite, garnet, diamond xenocrysts and mantle xenoliths
- Lamproite: similar, but saturated in SiO_2
- Mantle-derived rocks

Deposit types – kimberlite pipes
Diatremes

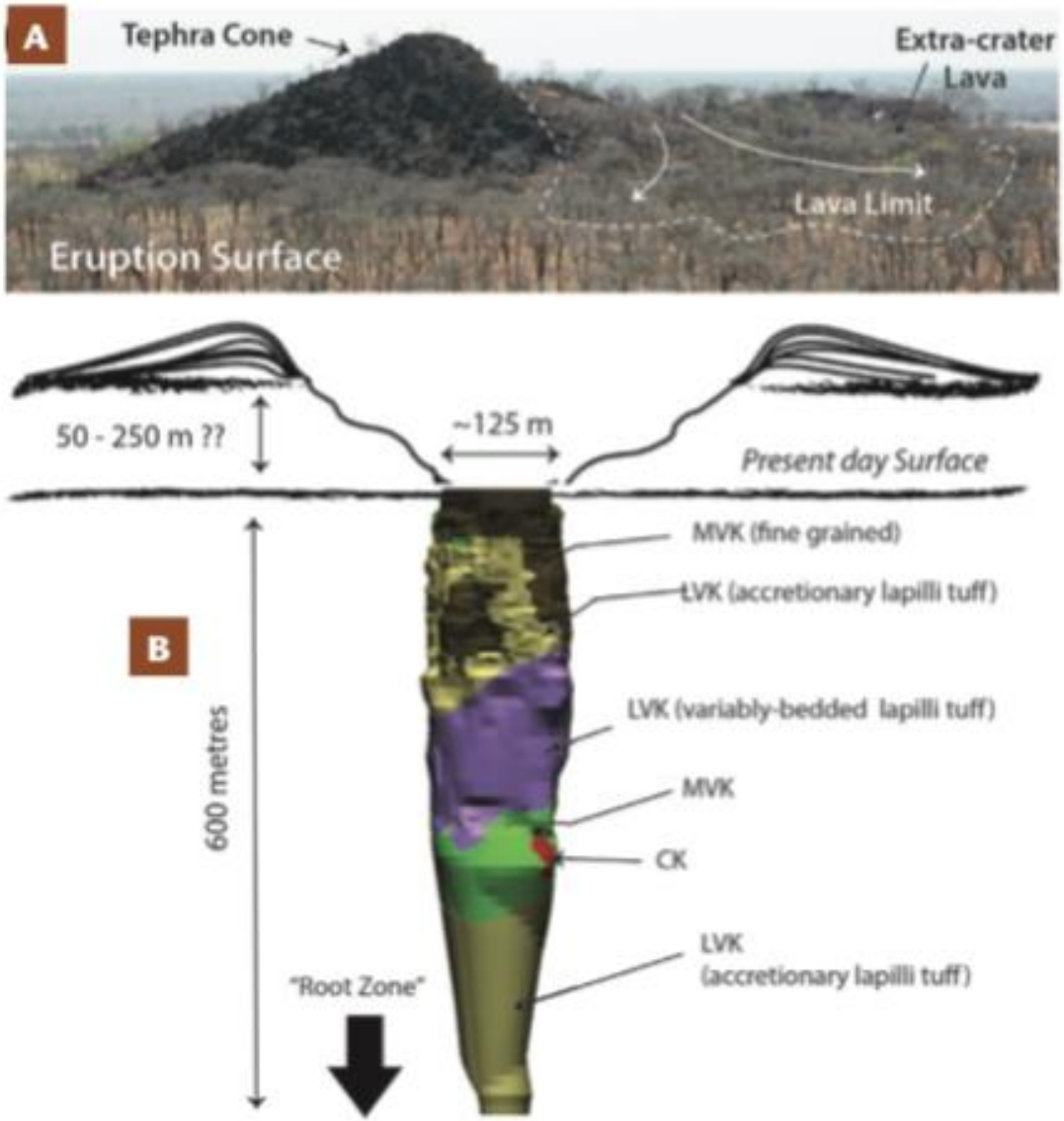


Pohl 2005

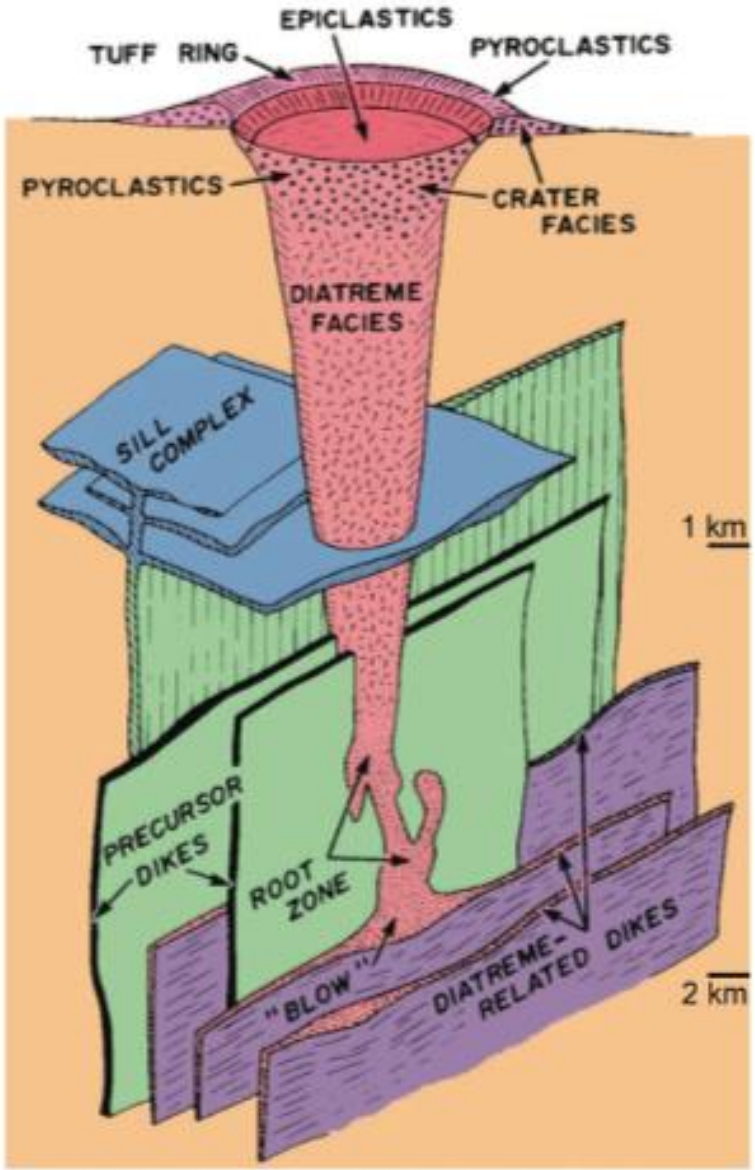



Premier Diamond Mine, South Africa. An illustration of extensive "open cut" mining

Deposit types – kimberlite pipes
Diatremes

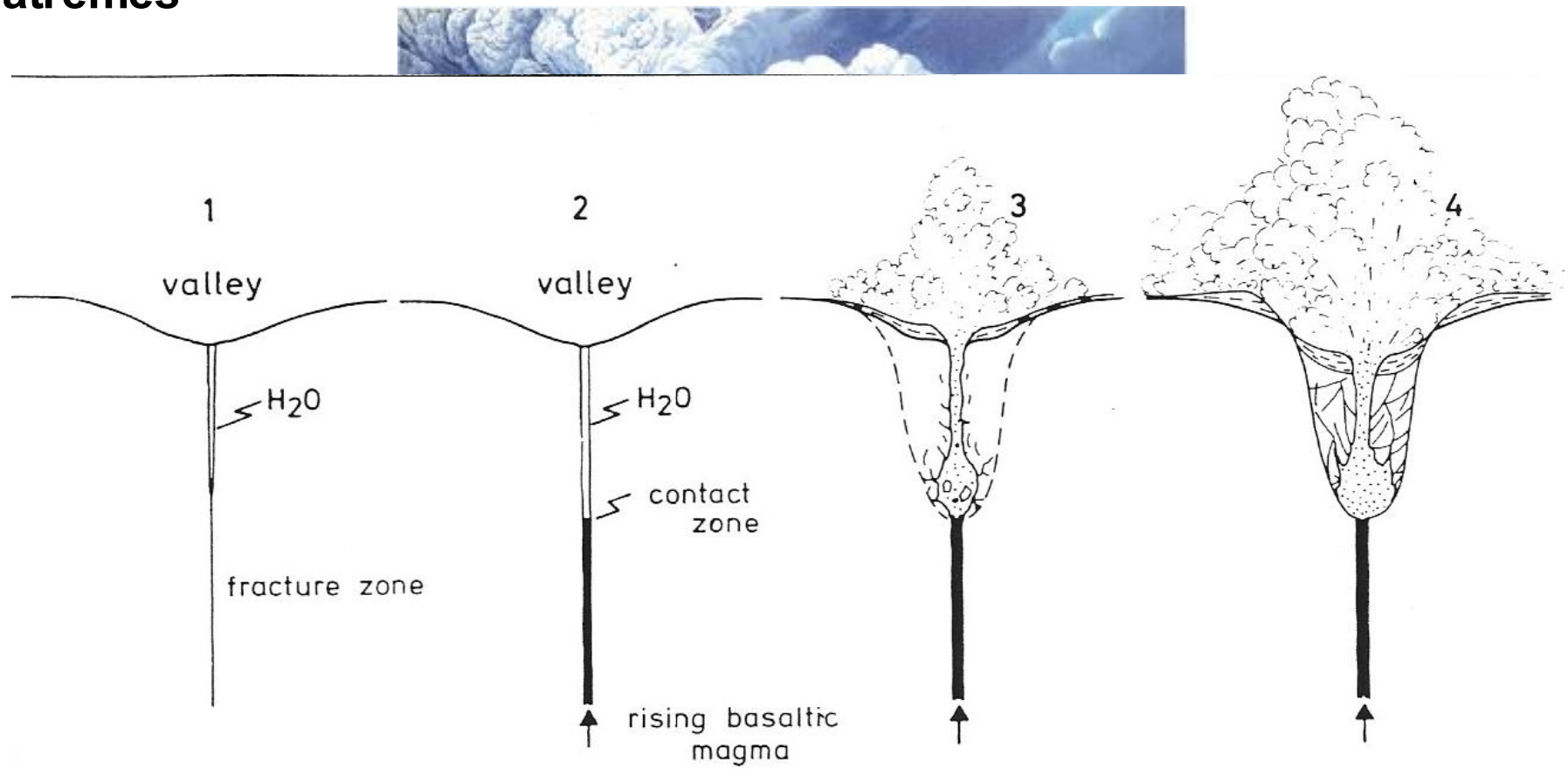


Russell 2019

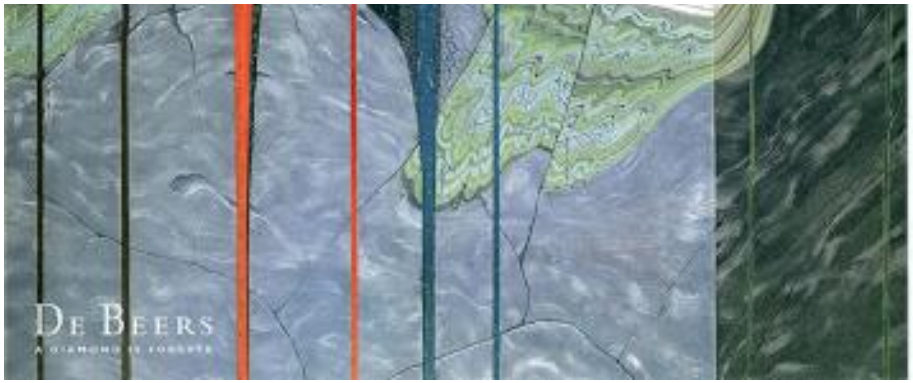


Mitchell 1986

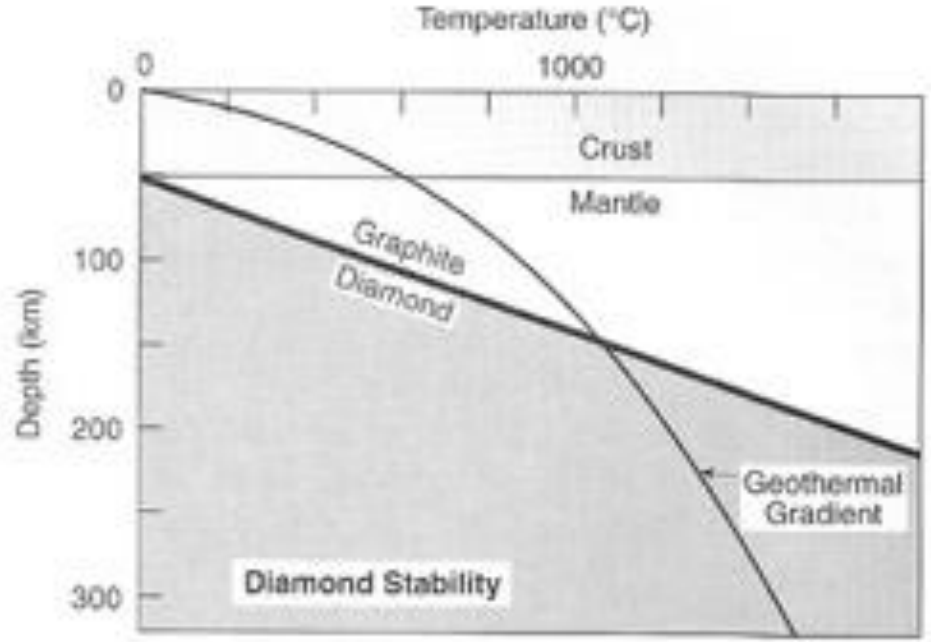
Deposit types – kimberlite pipes
Diatremes



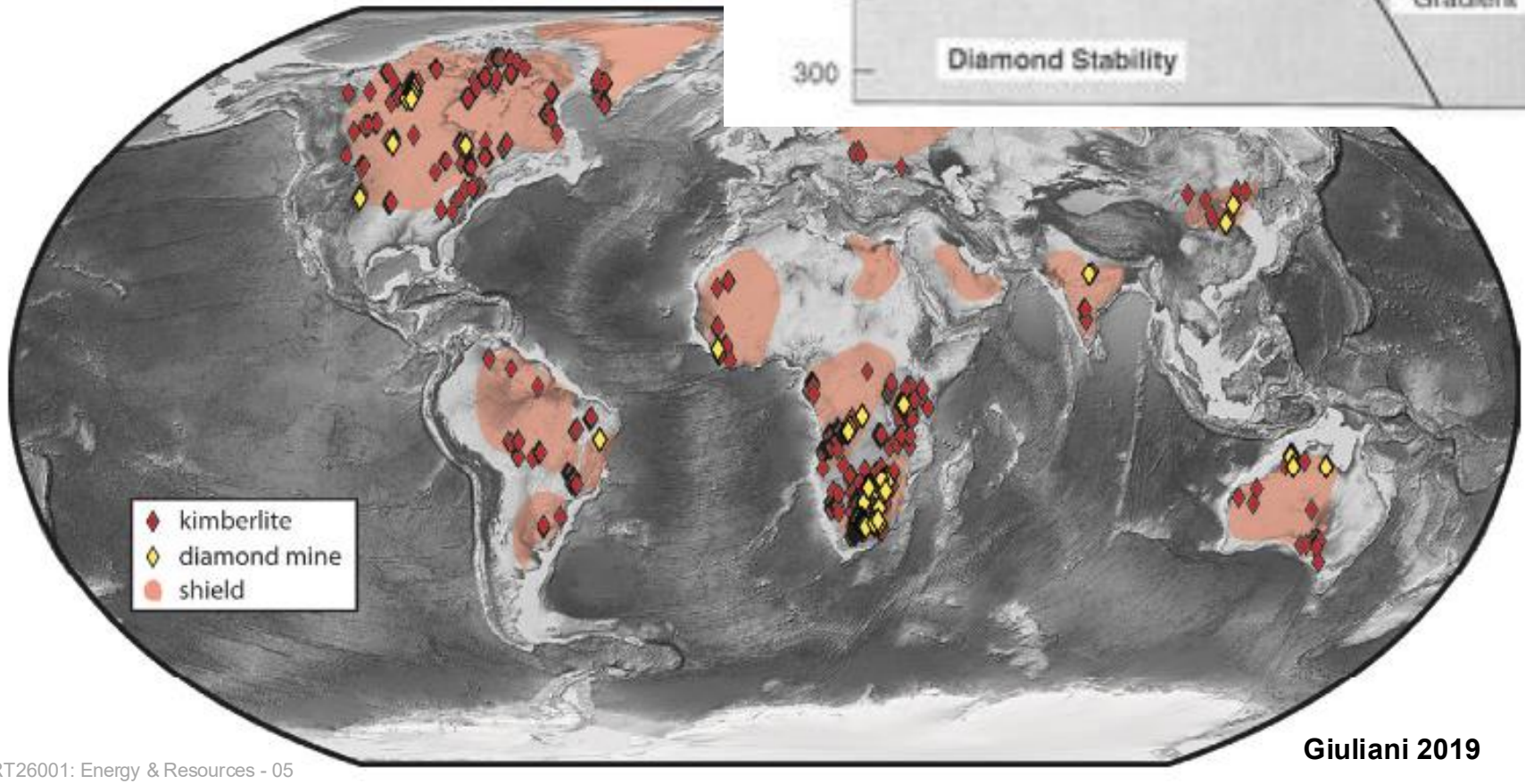
Lorenz 1986



Diamond genesis
Kimberlite distribution

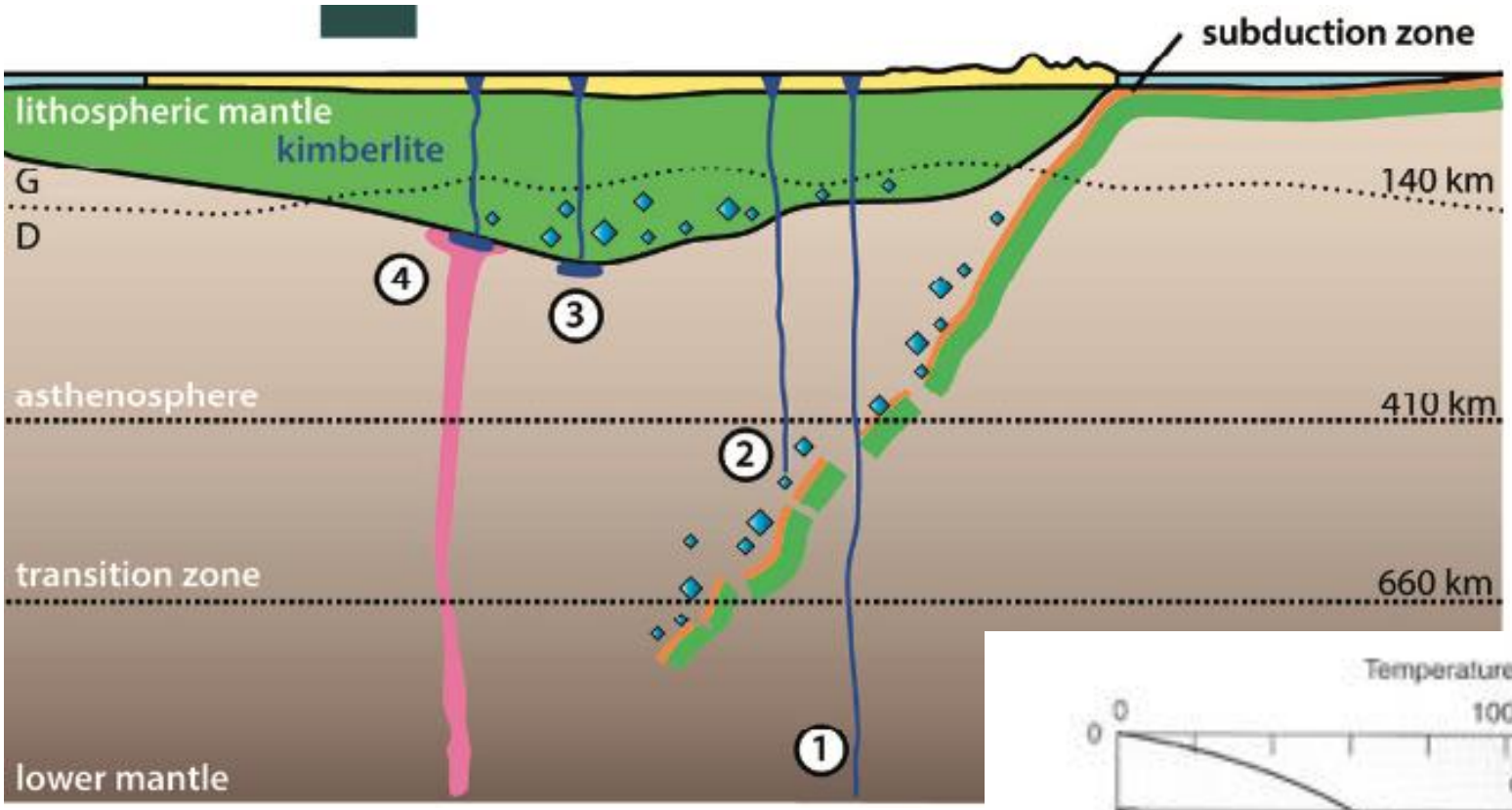


Kesler 1994

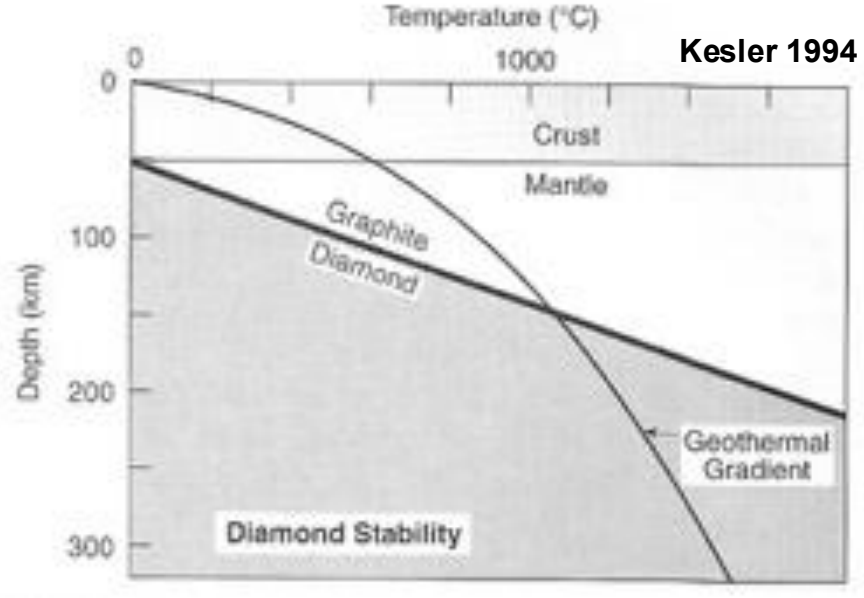


Giuliani 2019

Diamond genesis
Mantle processes

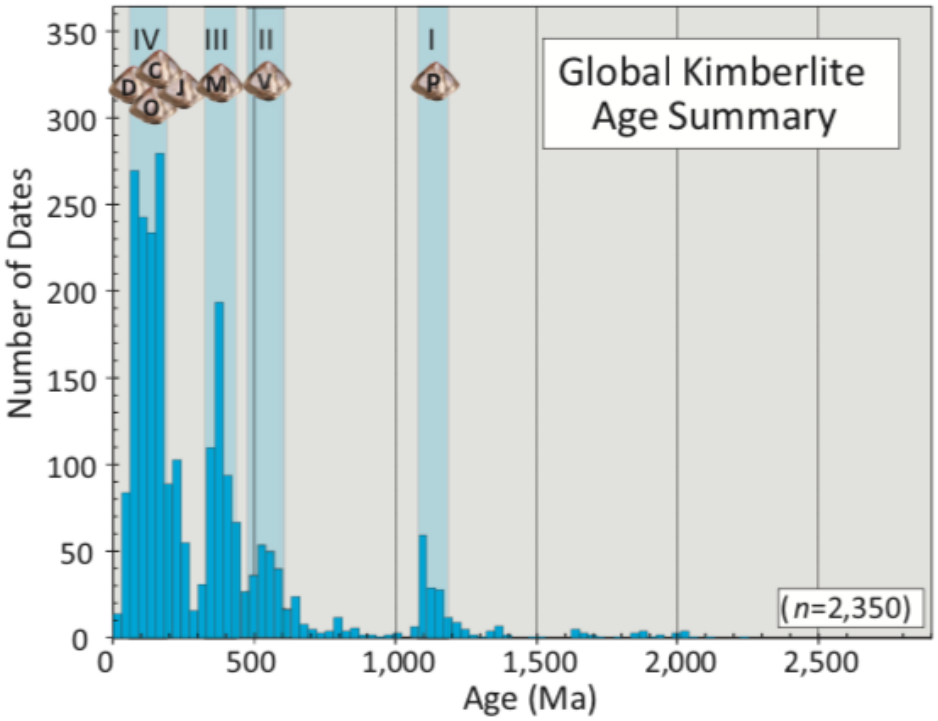


Giuliani 2019

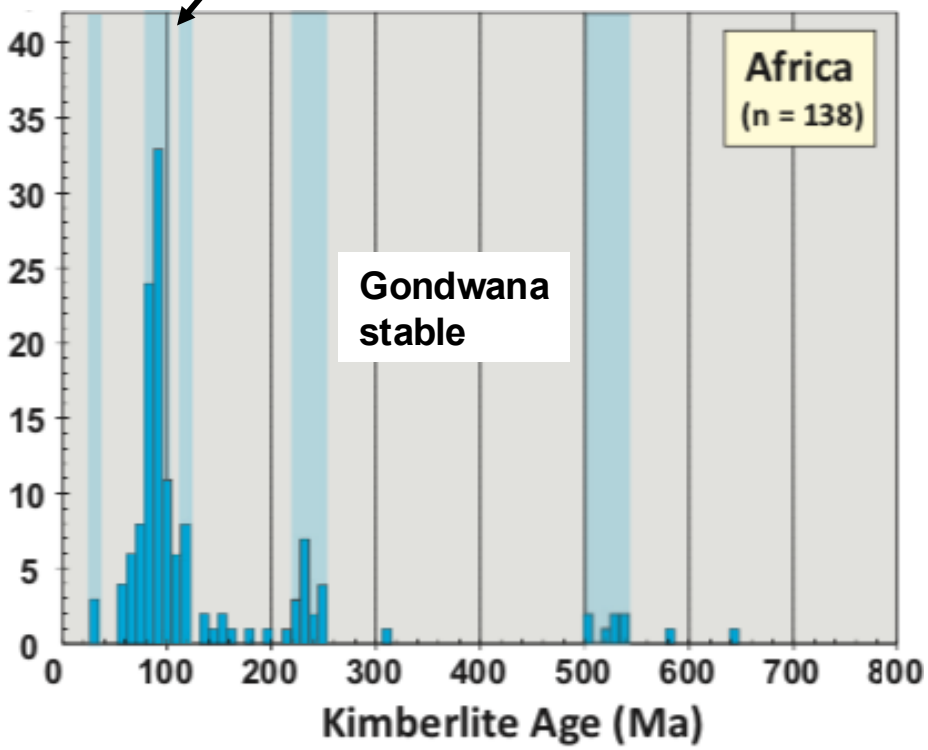


Kesler 1994

Diamond genesis
Global trends



Pangea rifting in S Atlantic?

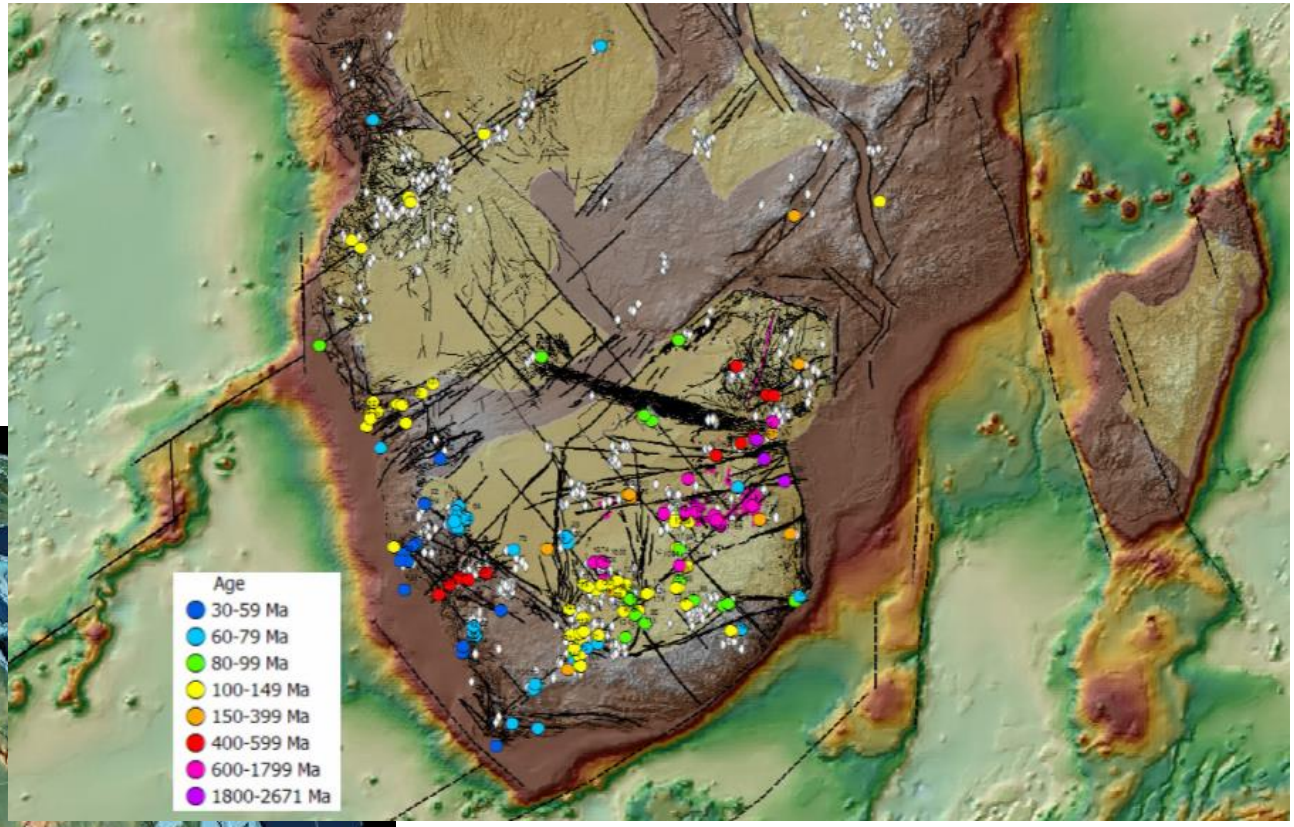


(in shaded light blue vertical bands). The age of kimberlites that host the world's largest diamond mines are shown with diamond symbols and identified by letters within these symbols: D = Diavik and Ekati Mines (Canada) (~48–56 Ma); O = Orapa Mine (Botswana) (93 Ma); C = Catoca Mine (Angola) (118 Ma); J = Jwaneng Mine (Botswana) (235 Ma); M = Mir Mine (Russia) (365 Ma); V = Venetia Mine (South Africa) (528 Ma); P= Premier pipe of the Cullinan Mine (South Africa) (1.2 Ga).

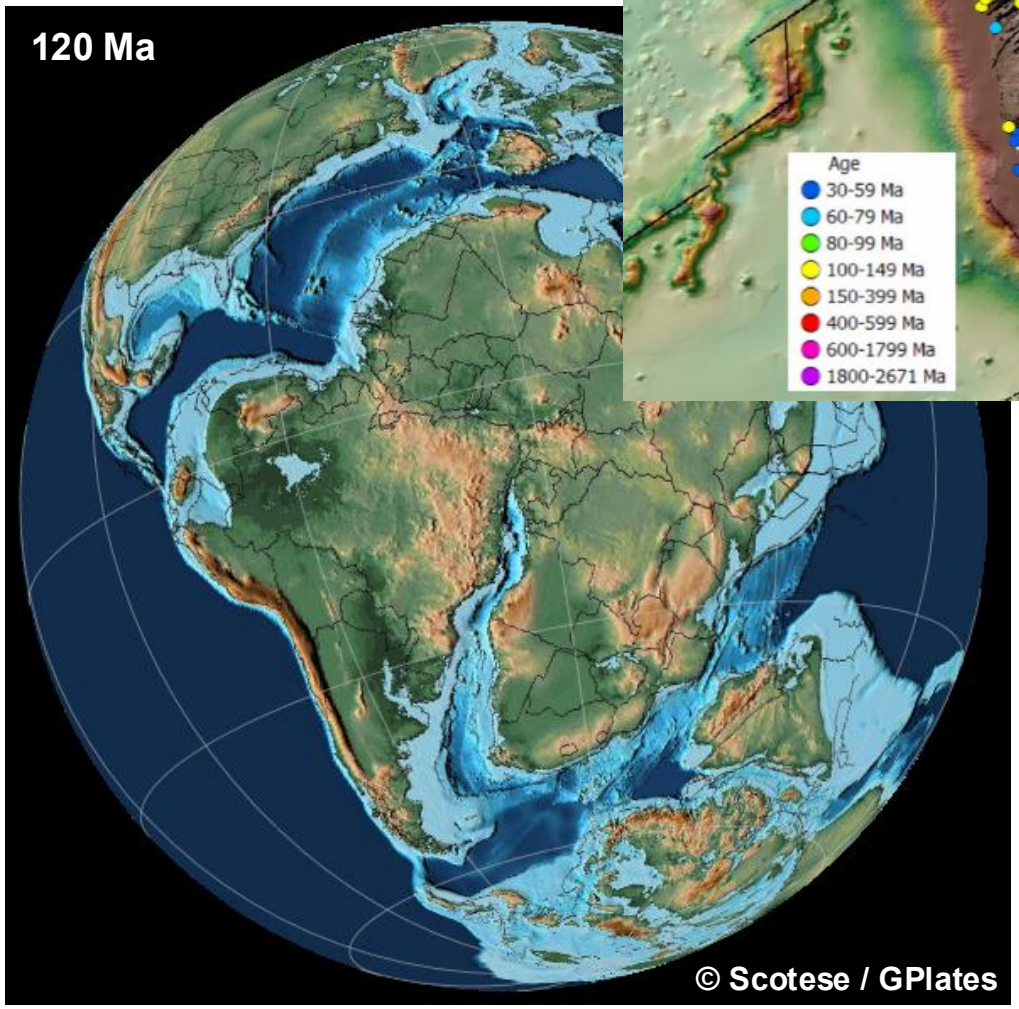
Heaman 2019

- **Kimberlite age distribution shows 'bar code' that may reflect periods of plate reconfiguration: rifting, subduction, mantle plumes**

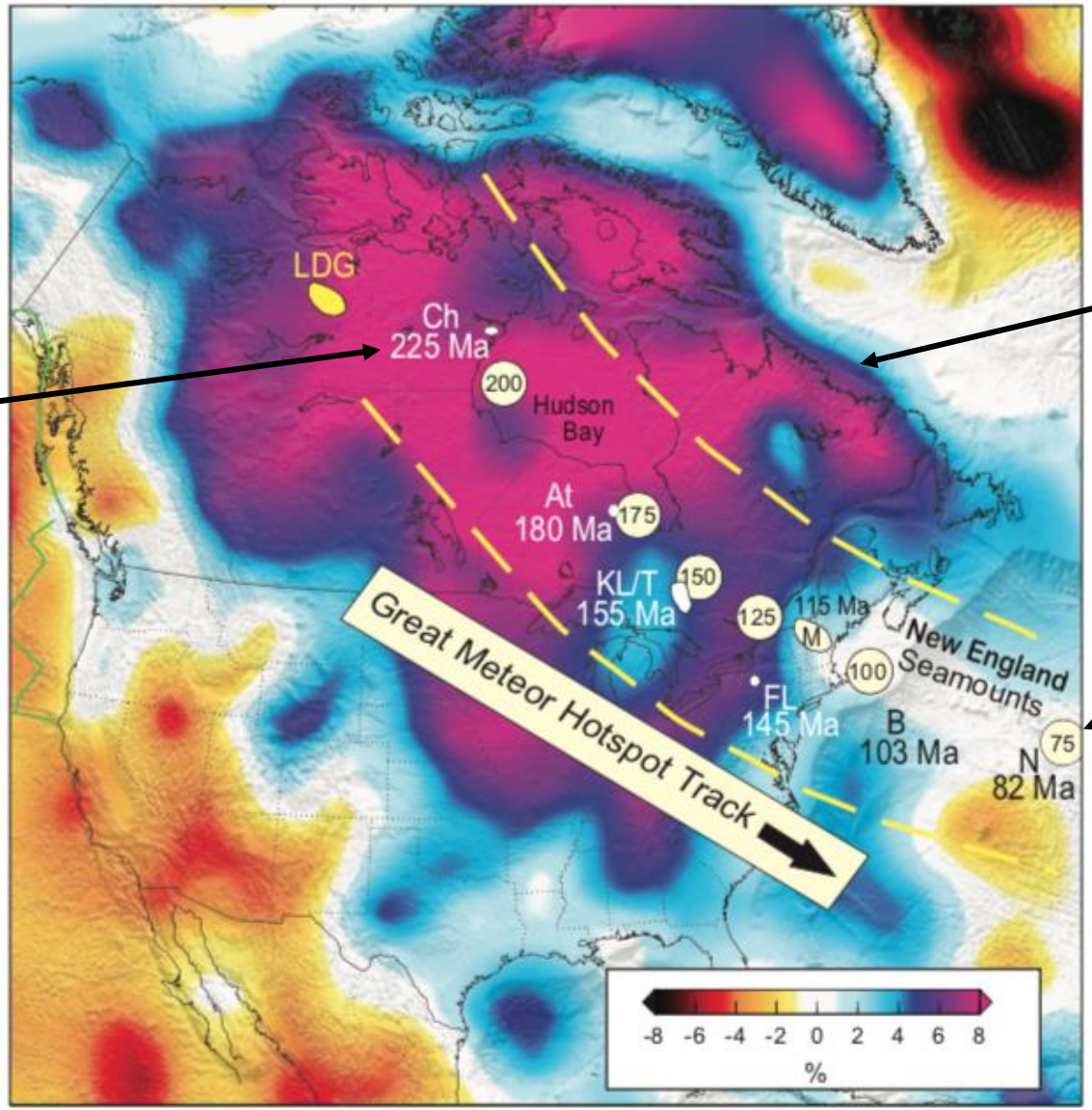
Diamond genesis
Global trends



Jelsma 2008



Diamond genesis
Global trends



Kimberlite fields and age

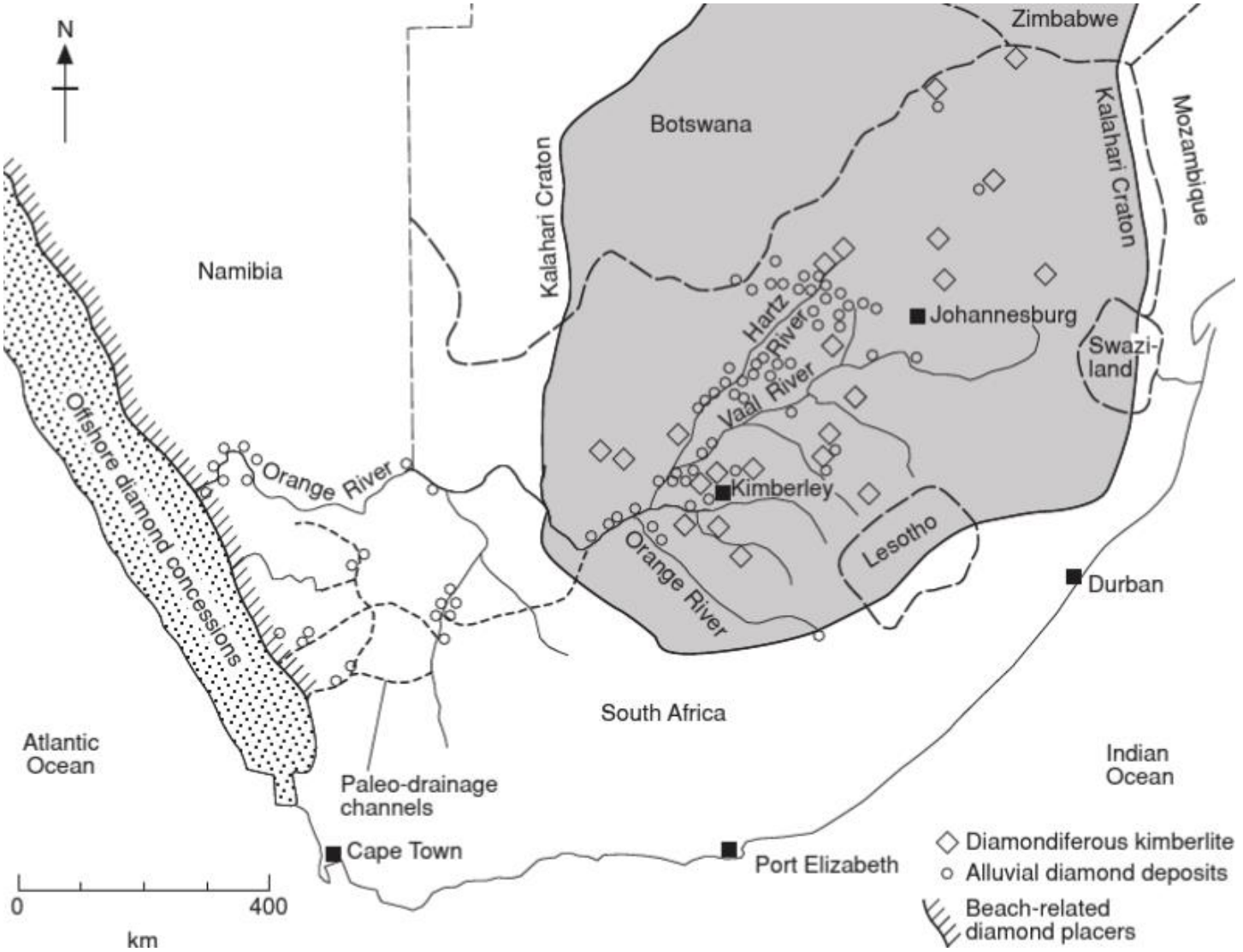
Thickness of lithosphere

Age track of plume

Heaman 2019

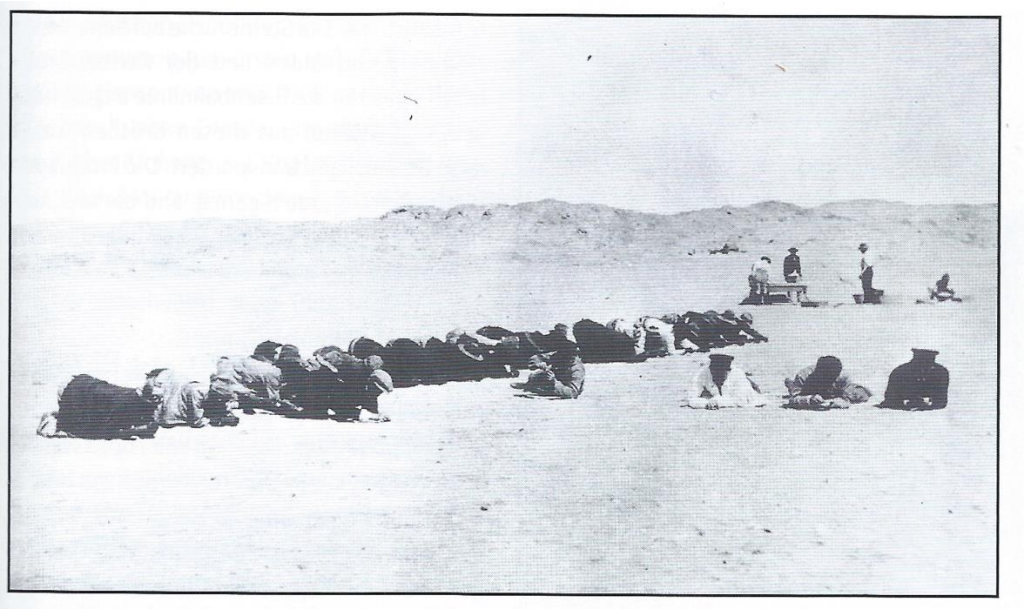
Deposit types – placer deposits
Distribution

Distribution of primary and placer deposits in southern Africa



Robb 2005

Deposit types – placer deposits
Distribution



Schneider 2000

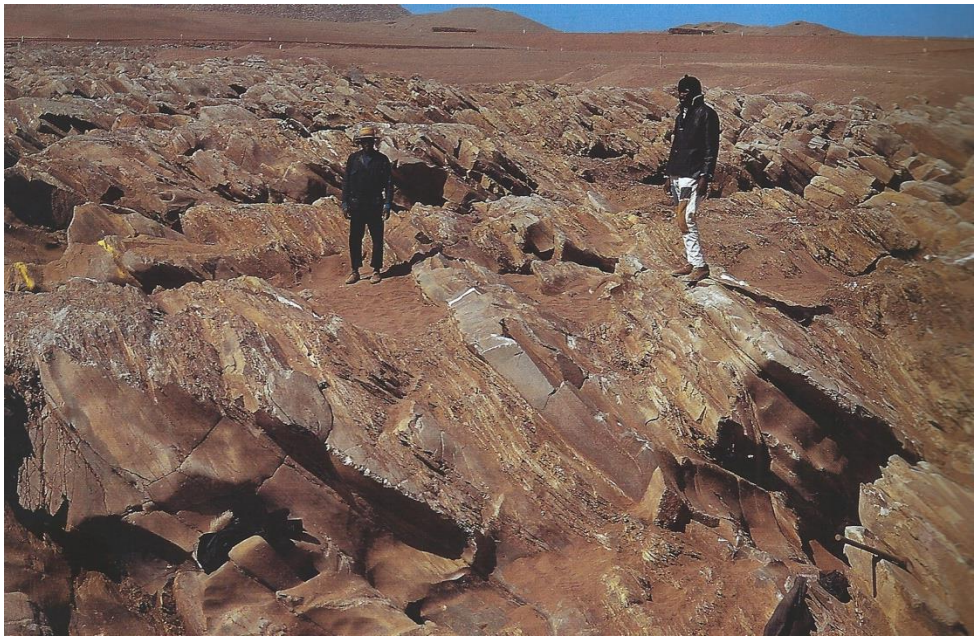


© Lucapa diamonds

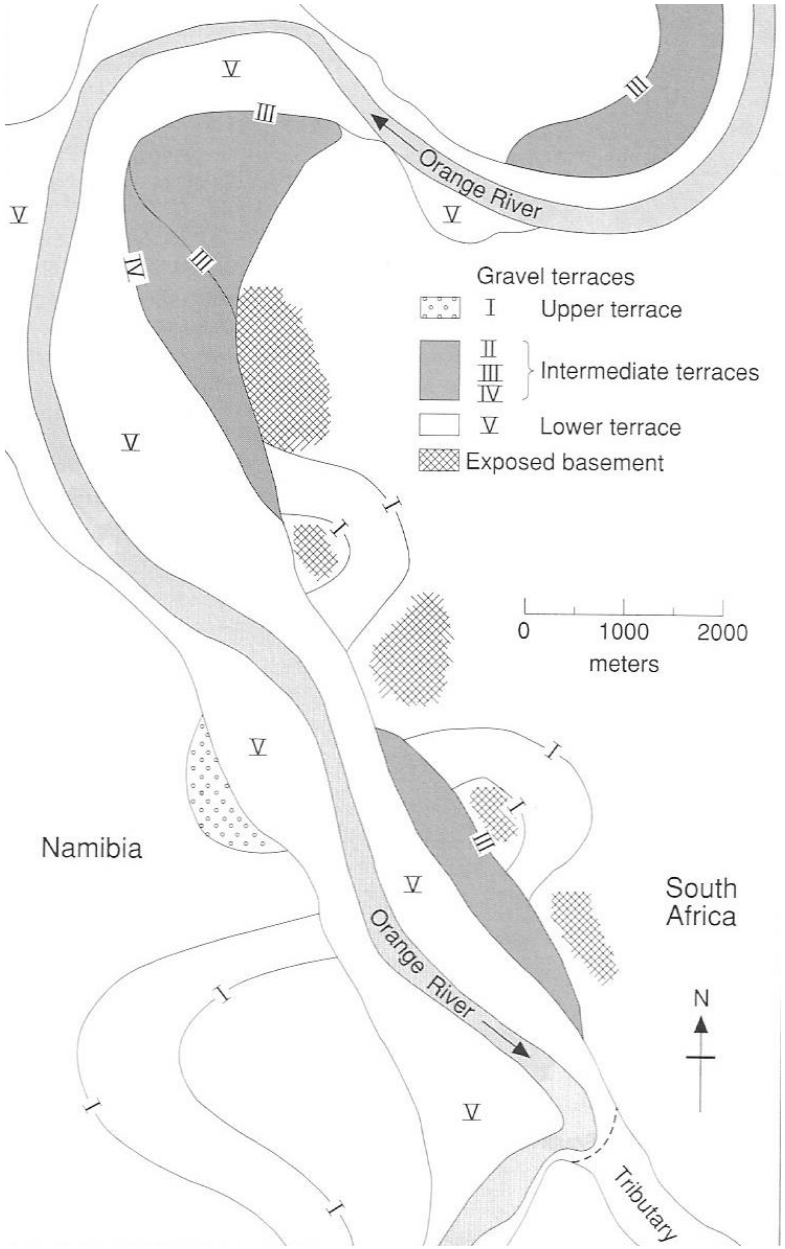


Schneider 2000

Deposit types – placer deposits
Processes for concentration

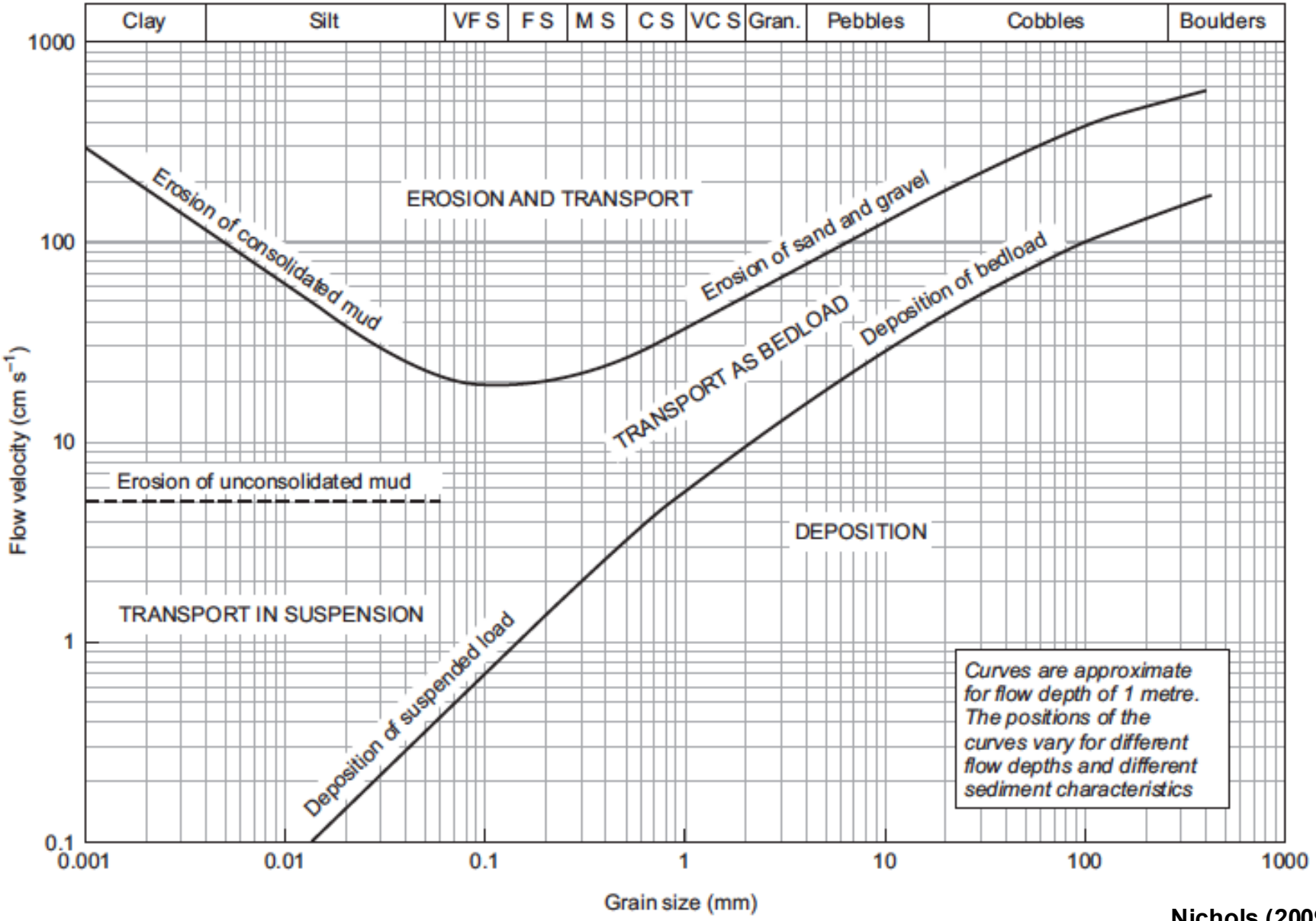


Schneider 2000



Robb 2005

Deposit types – placer deposits
Processes for concentration



Nichols (2009)

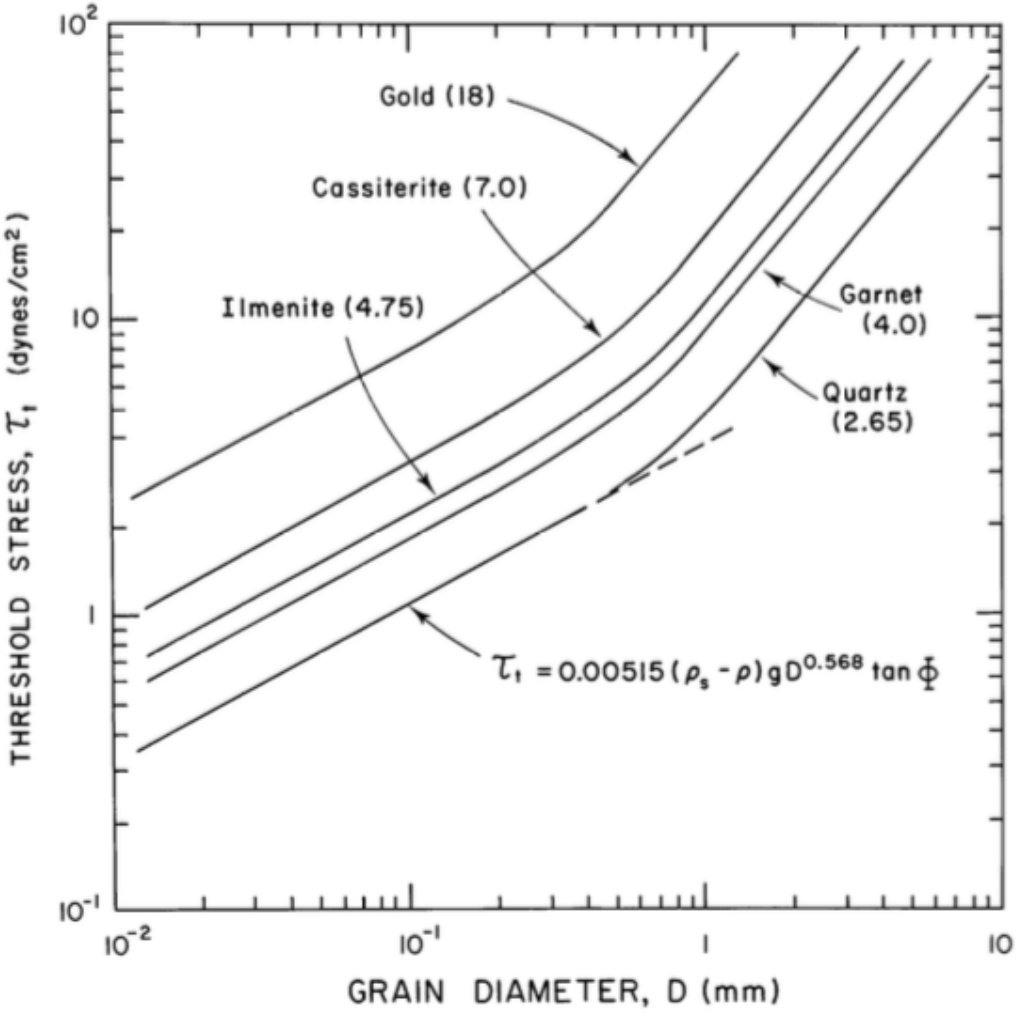
Deposit types – placer deposits
Processes for concentration



© Prospecting and Mining Journal

$$V = \frac{(\rho_1 - \rho) \times g \times d^2}{18 \times \mu}$$

- Accumulation due to density and resistance to abrasion
- Accumulation in pockets in basement (potholes)

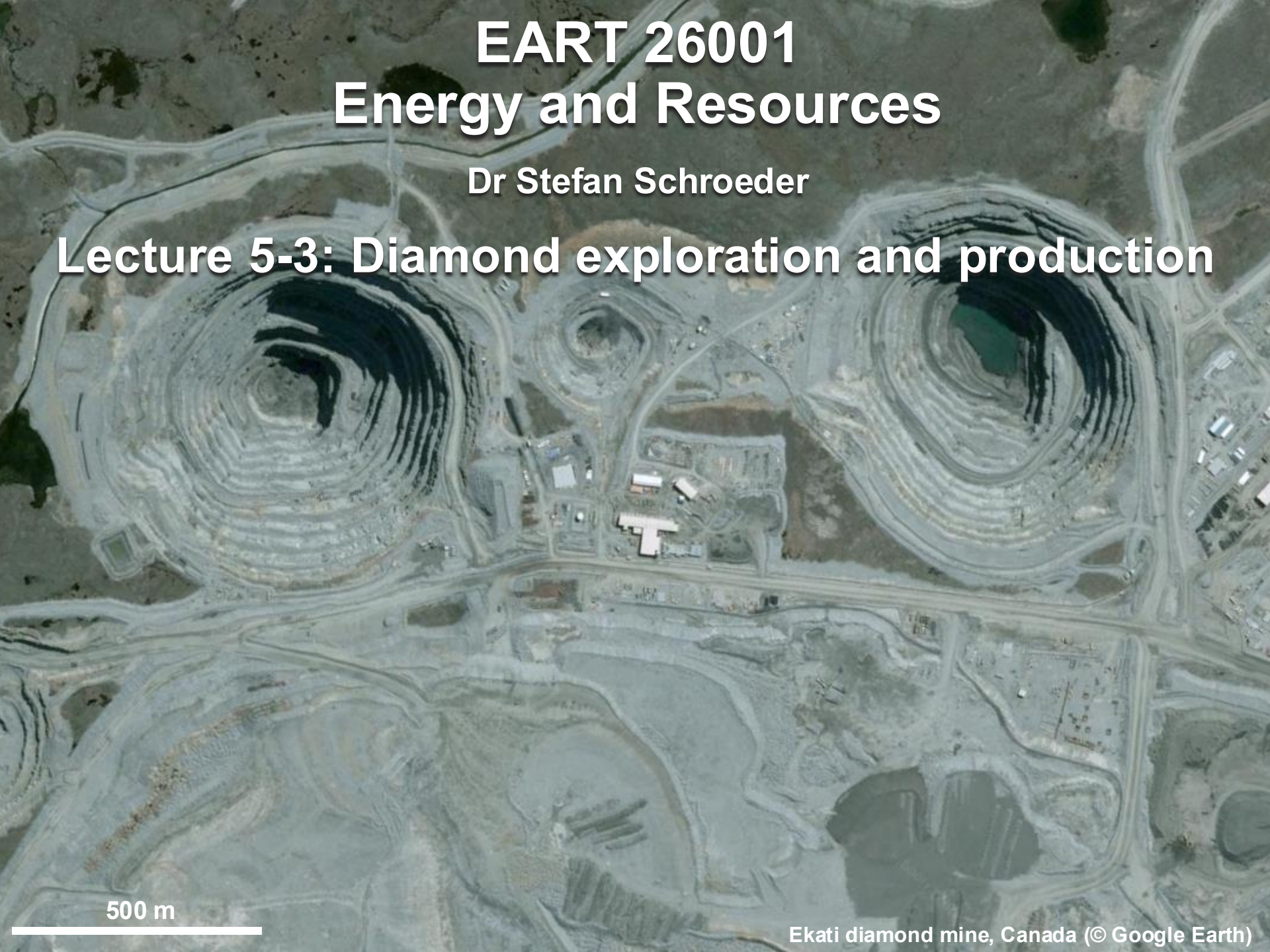


Komar 2007

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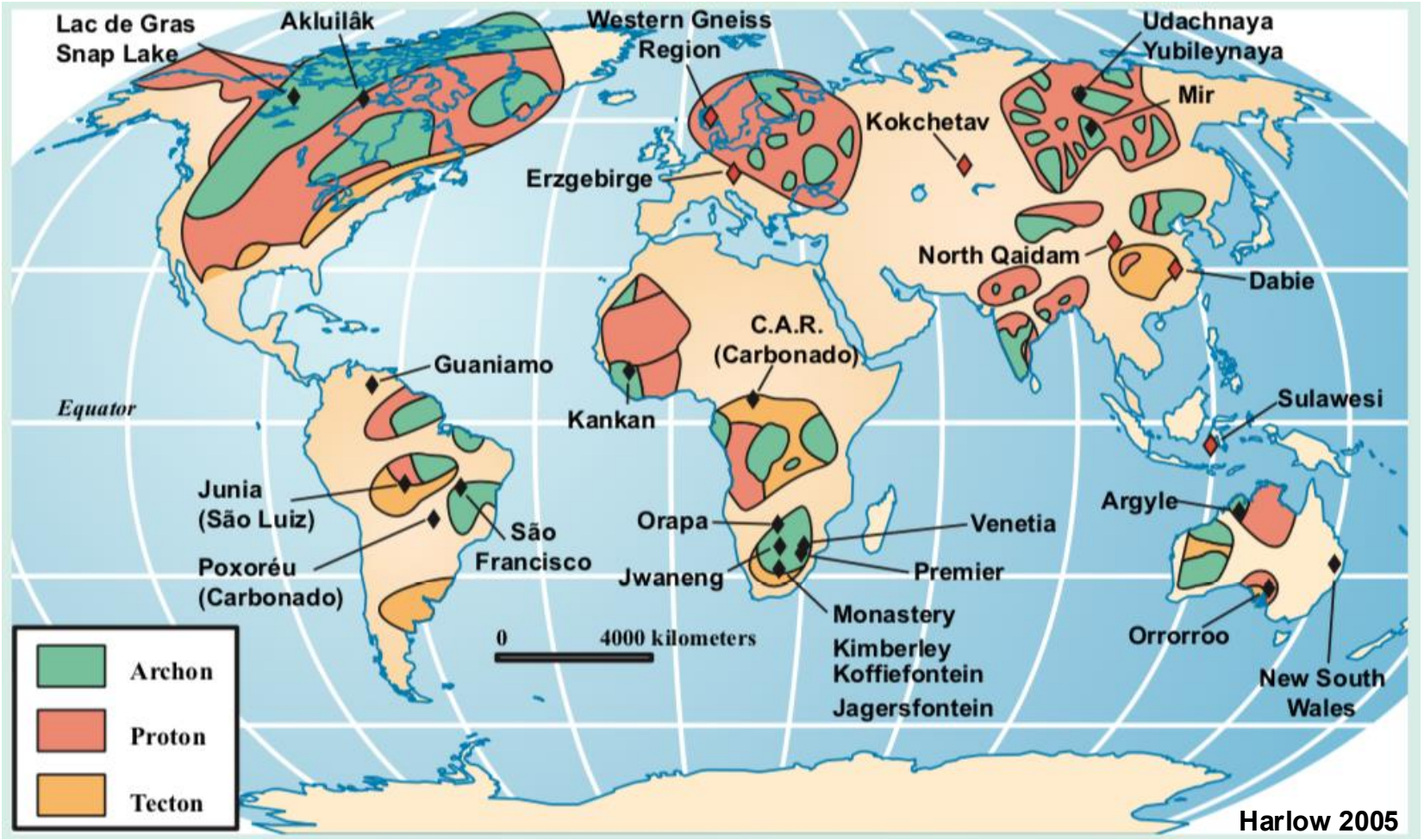
Dr Stefan Schroeder

Lecture 5-3: Diamond exploration and production



Ekati diamond mine, Canada (© Google Earth)

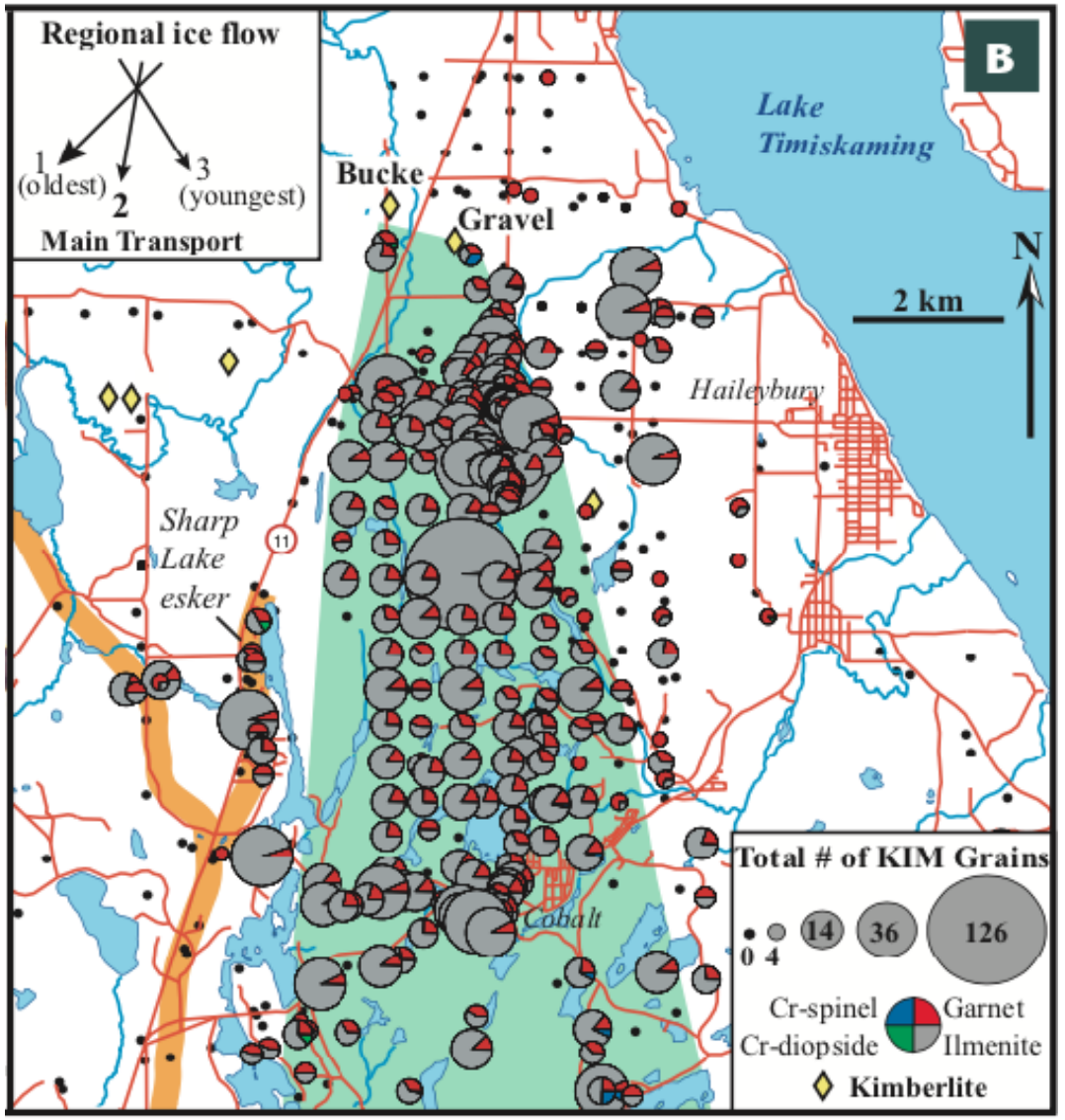
Exploration
Structural context



- Understand regional and local structural context
- Predict lithospheric regions where (and when) diamonds formed and remained stable, and understand processes that brought them to surface

Exploration
Indicator mineral sampling



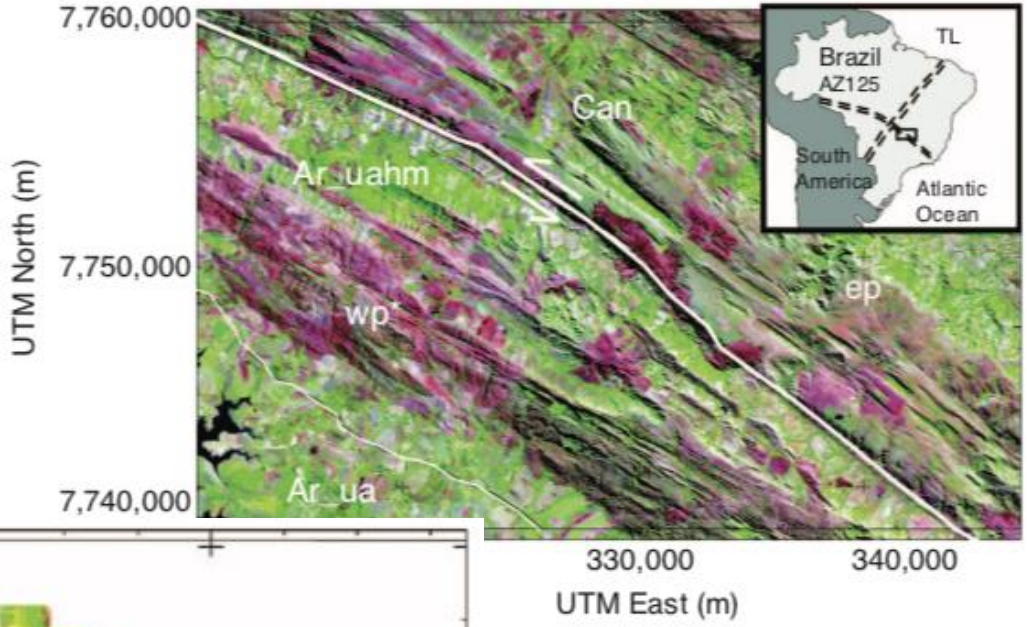


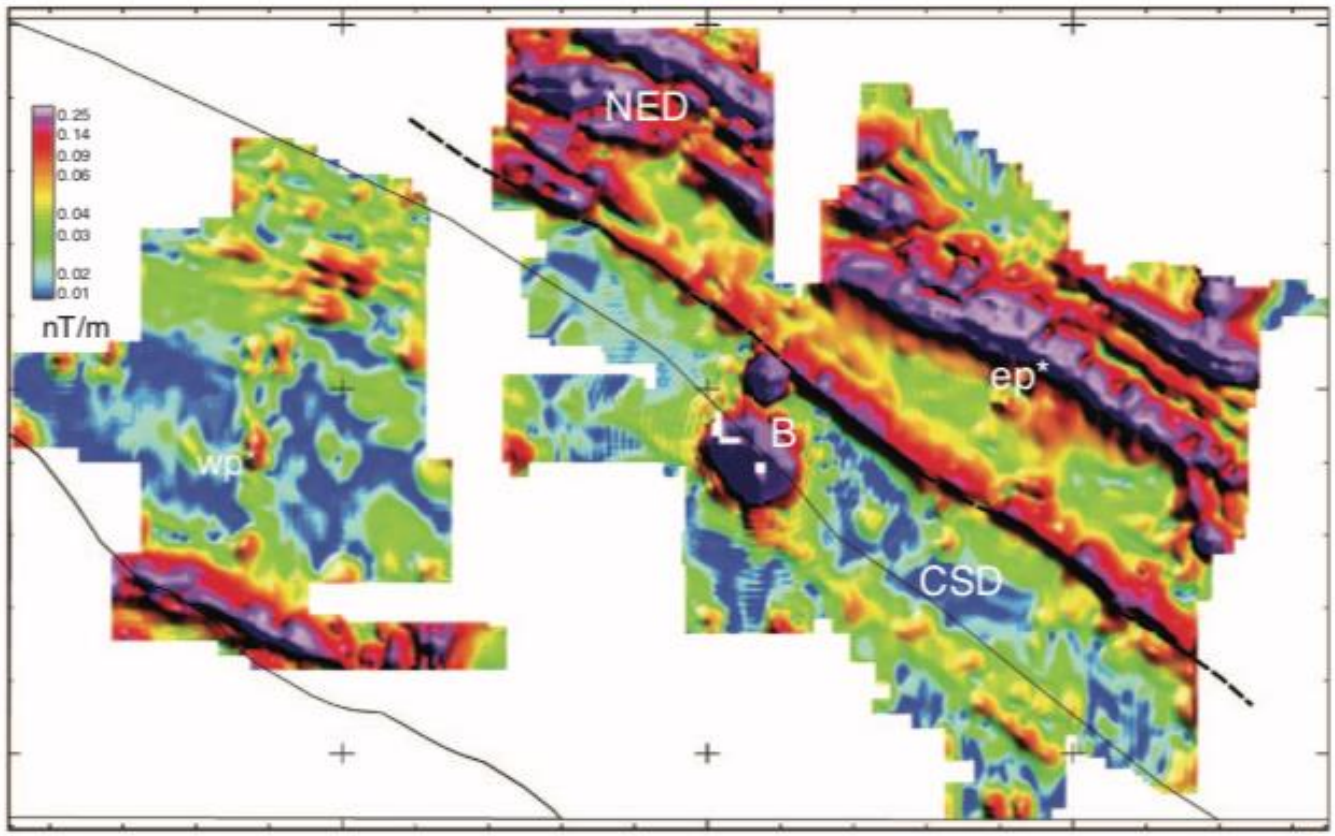
Kjarsgaard 2019

- Analysis of heavy mineral placers directly associated with kimberlite
- Landscape evolution

Exploration
Geophysics

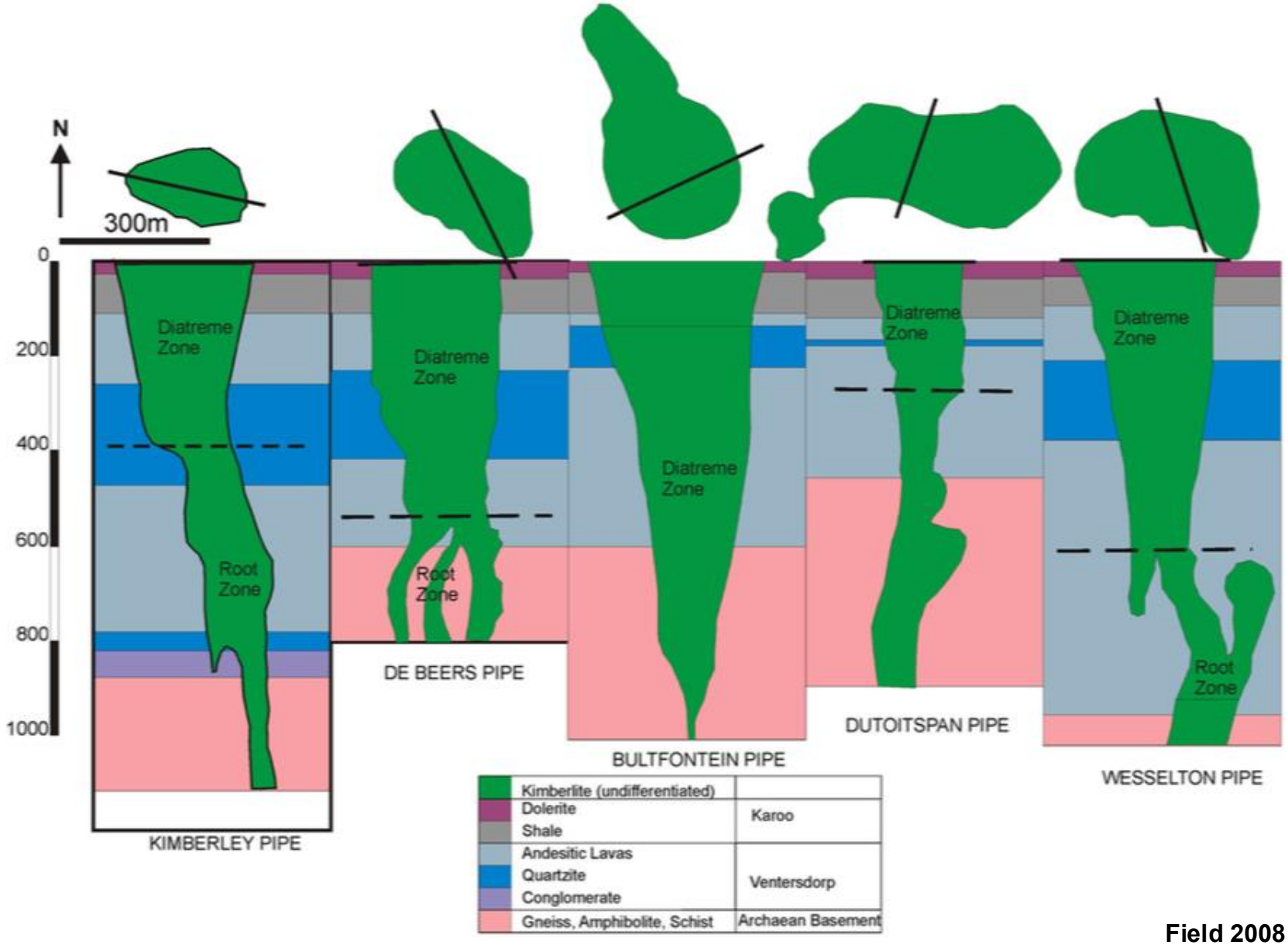
- **Airborne magnetics: difference in magnetic susceptibility to country rock**
- **Density contrast pipe / country rock**
- **Resistivity contrast pipe / country rock**





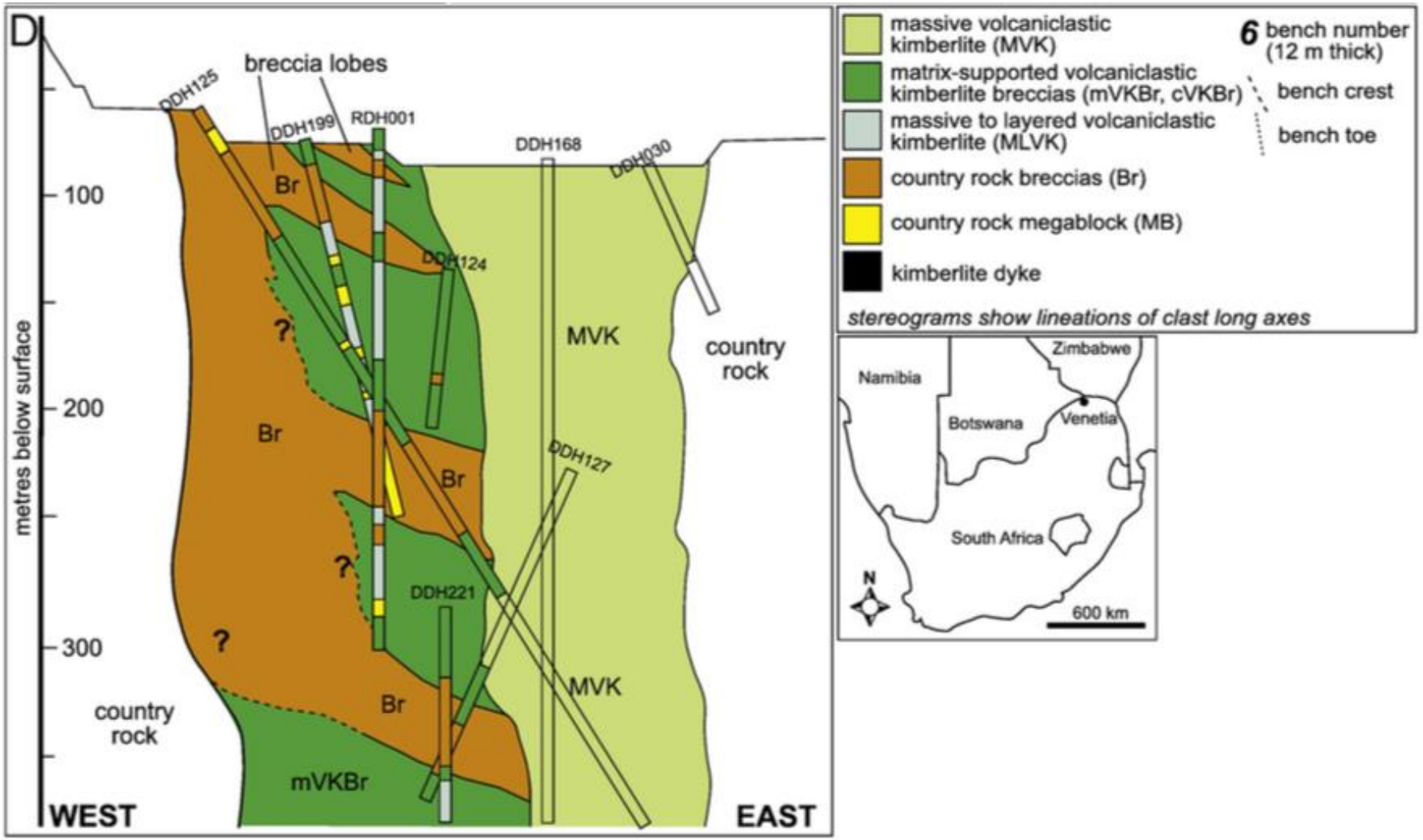
Menezes 2007

Exploration
Drilling and pipe models



Field 2008

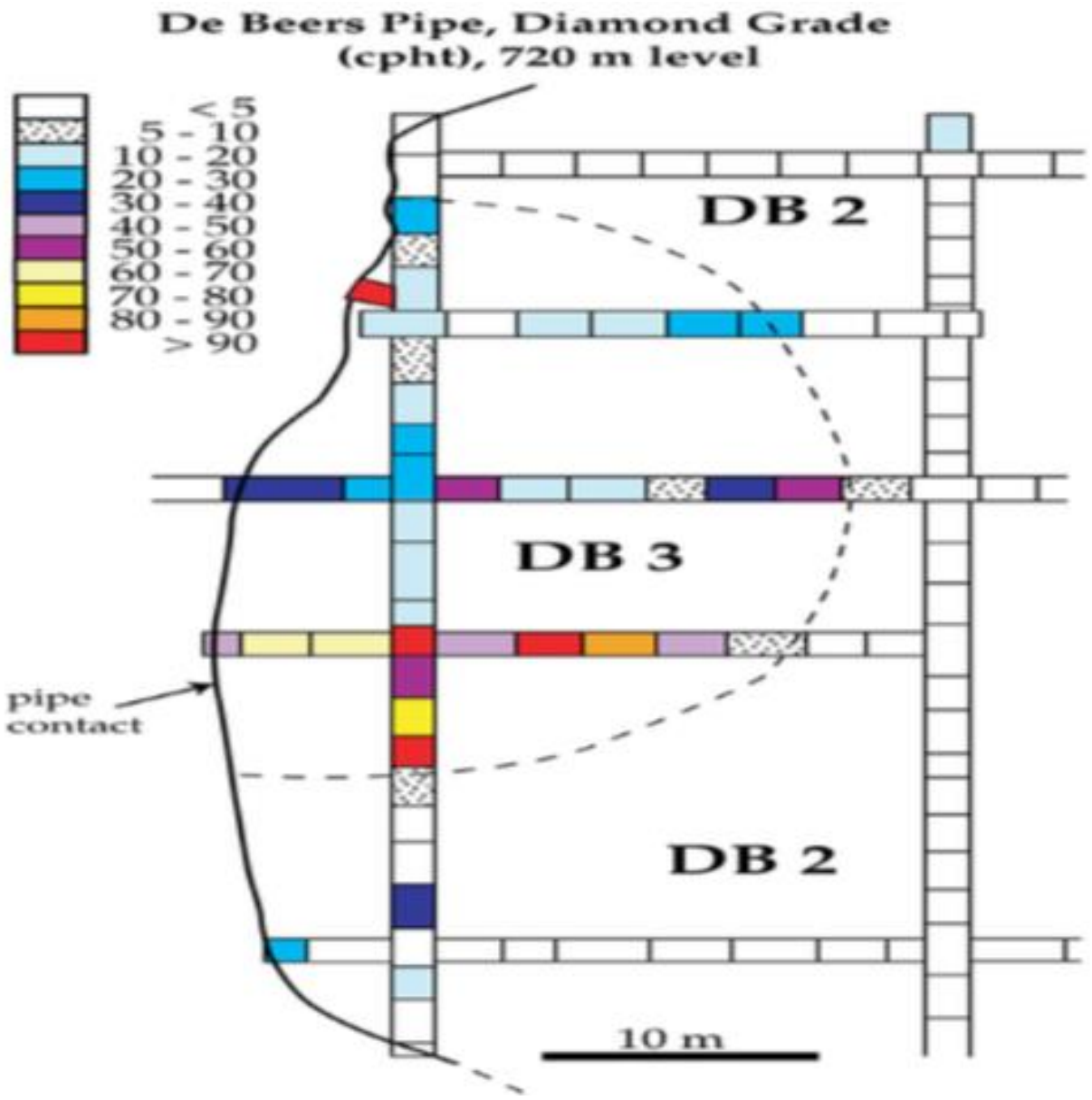
Exploration
Drilling and pipe models



Field 2008

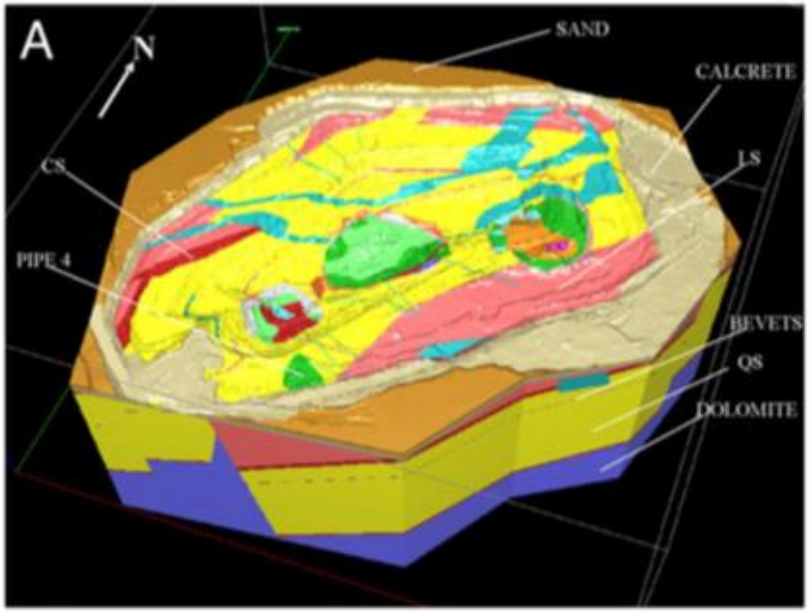
Exploration
Drilling and pipe models

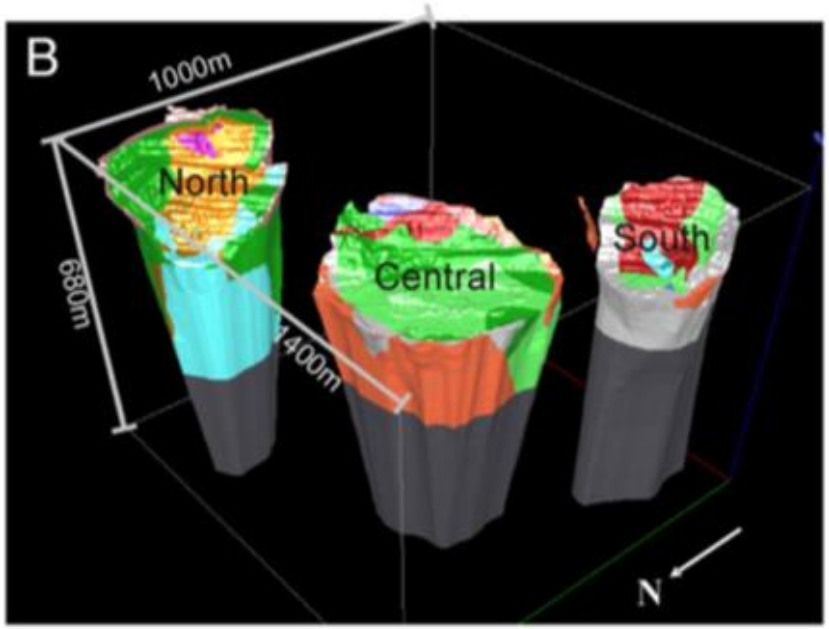
- Combination of geophysical and drilling data to build 3D model of pipe
- Massive samples (>100 t rock) to evaluate mining potential

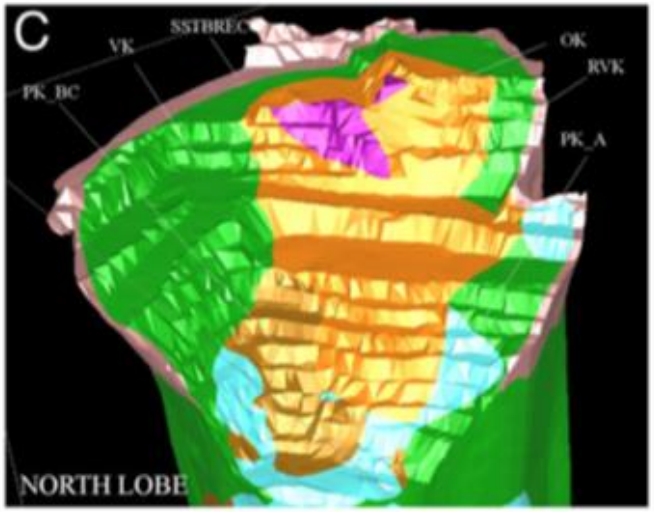


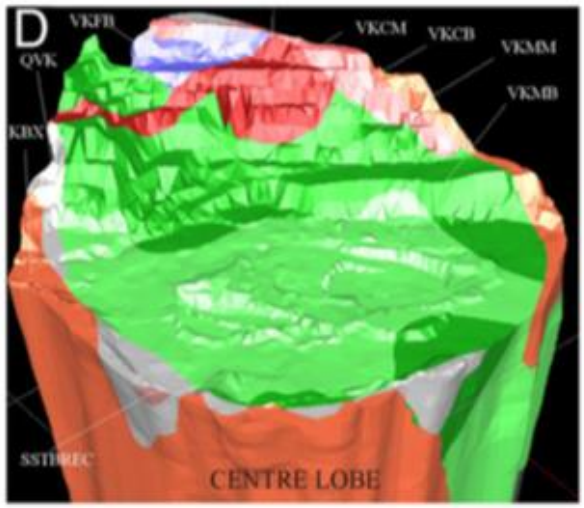
Kjarsgaard 2007

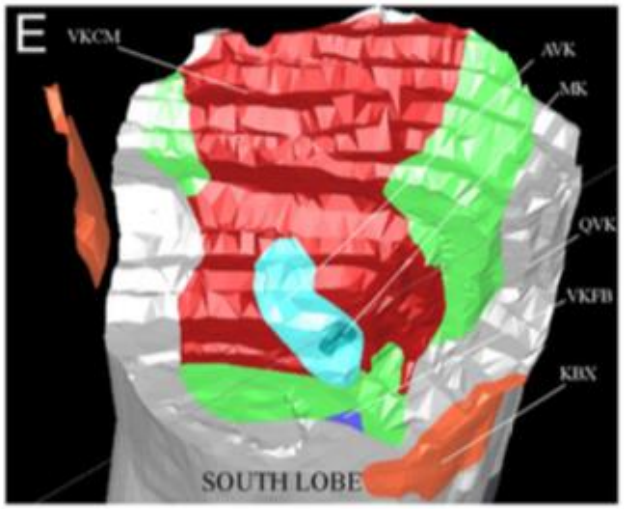
Exploration
Drilling and pipe models





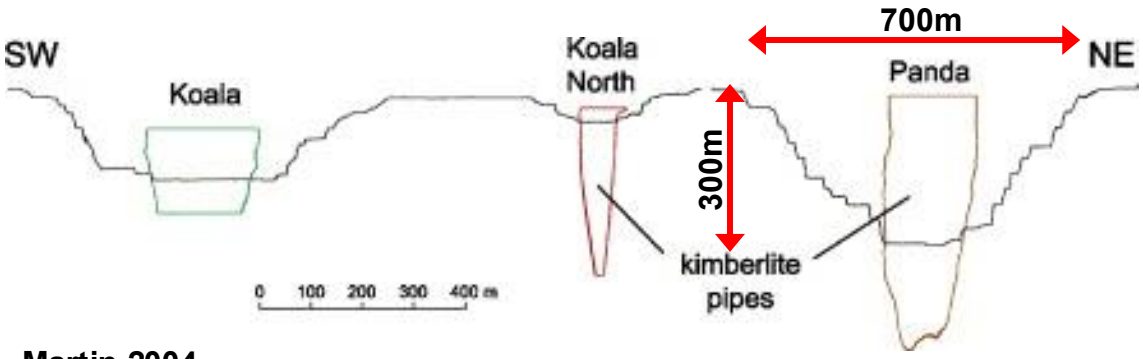






Field 2008

Production
Kimberlite pipes



Martin 2004

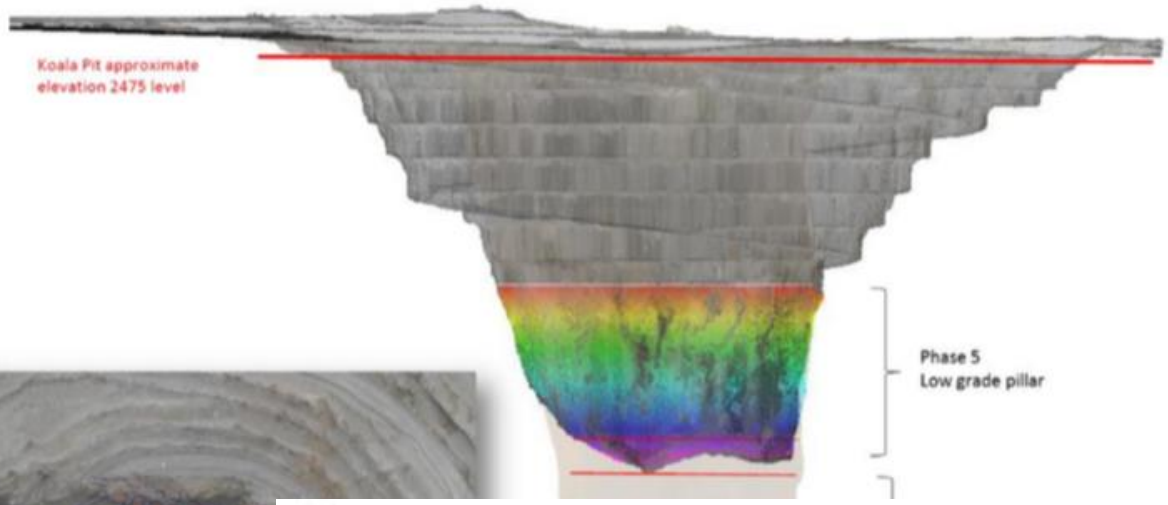


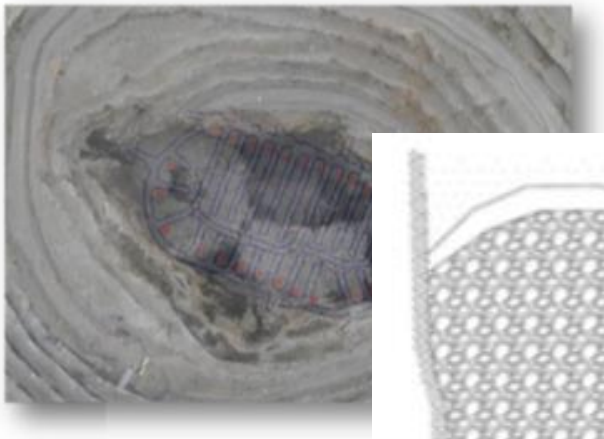
- Open pit blasting and excavating
- As long as economically and geotechnically feasible



Jakubec 2008

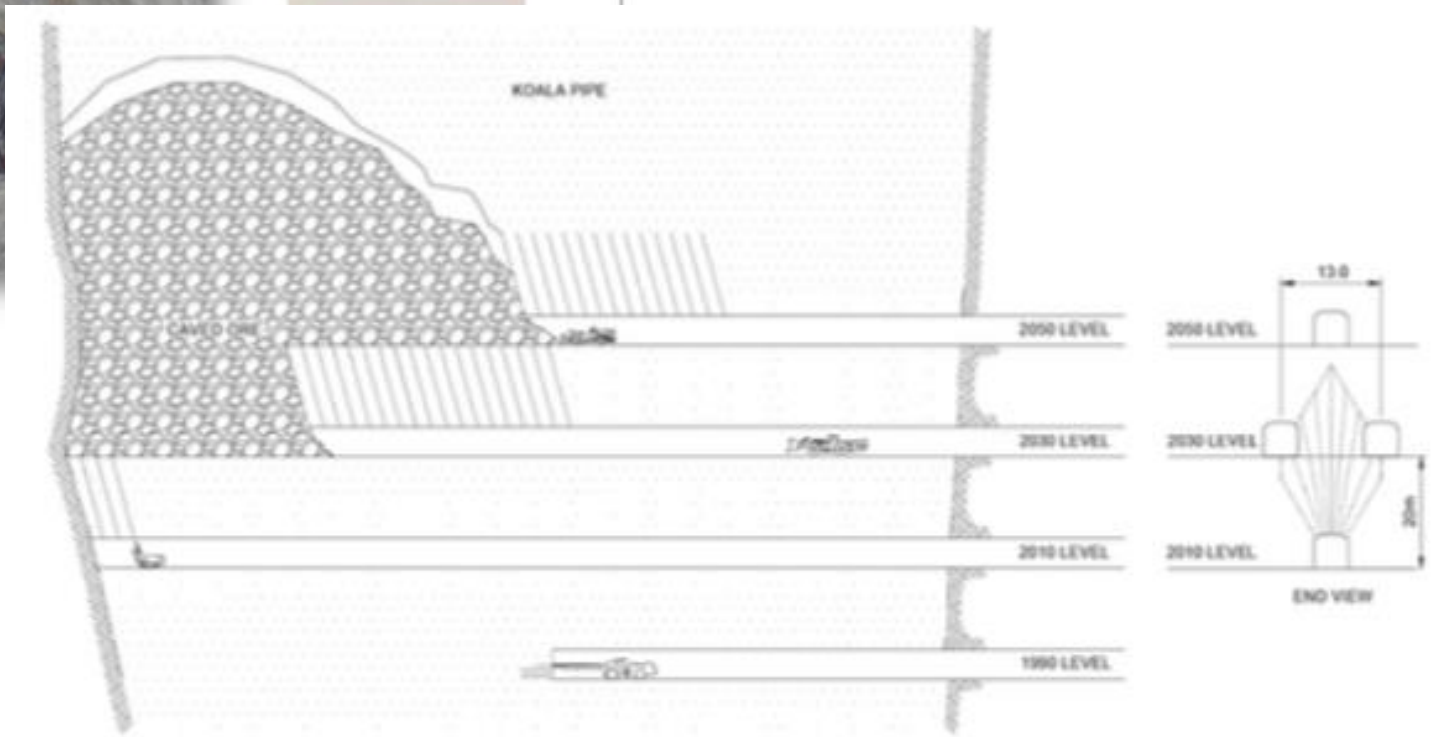
Production
Kimberlite pipes



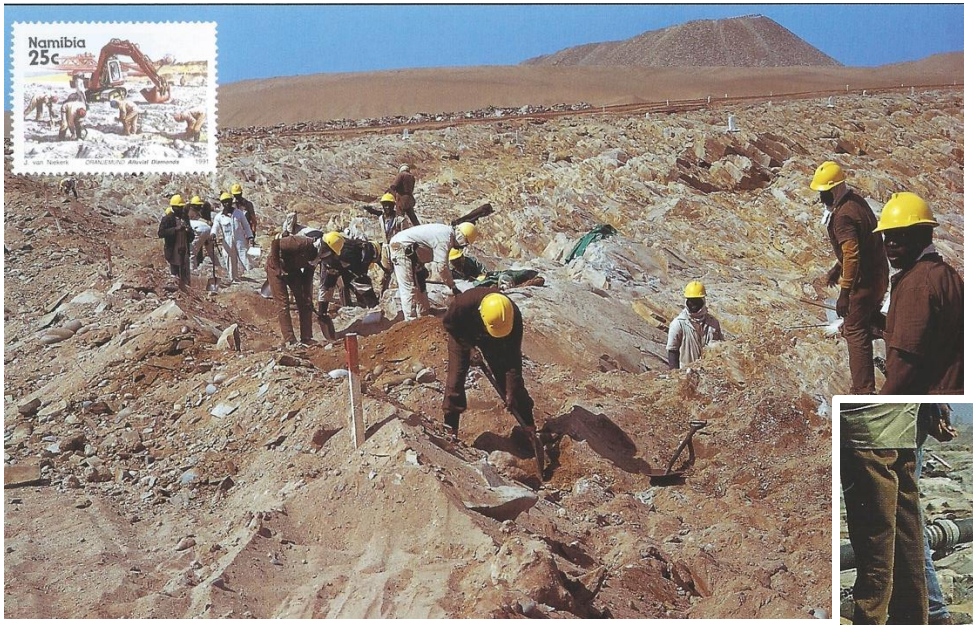


Jakubec 2008

- Underground mining
- Various techniques to blast and recover ore

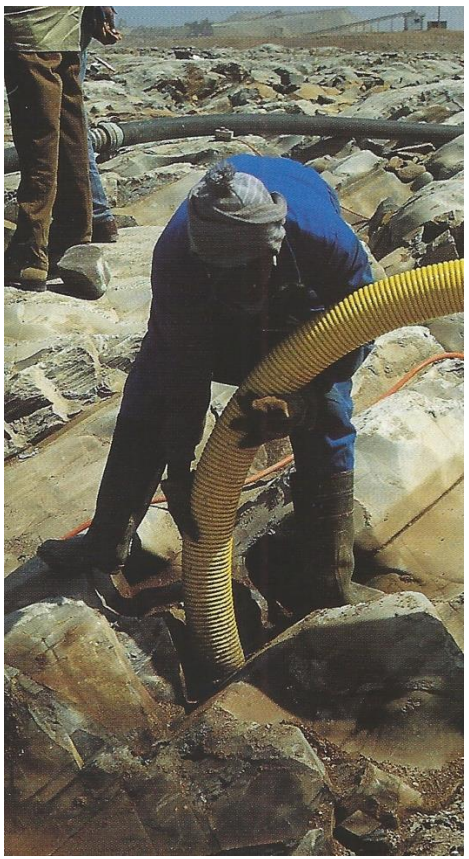


Production
Alluvial and marine diamond mines



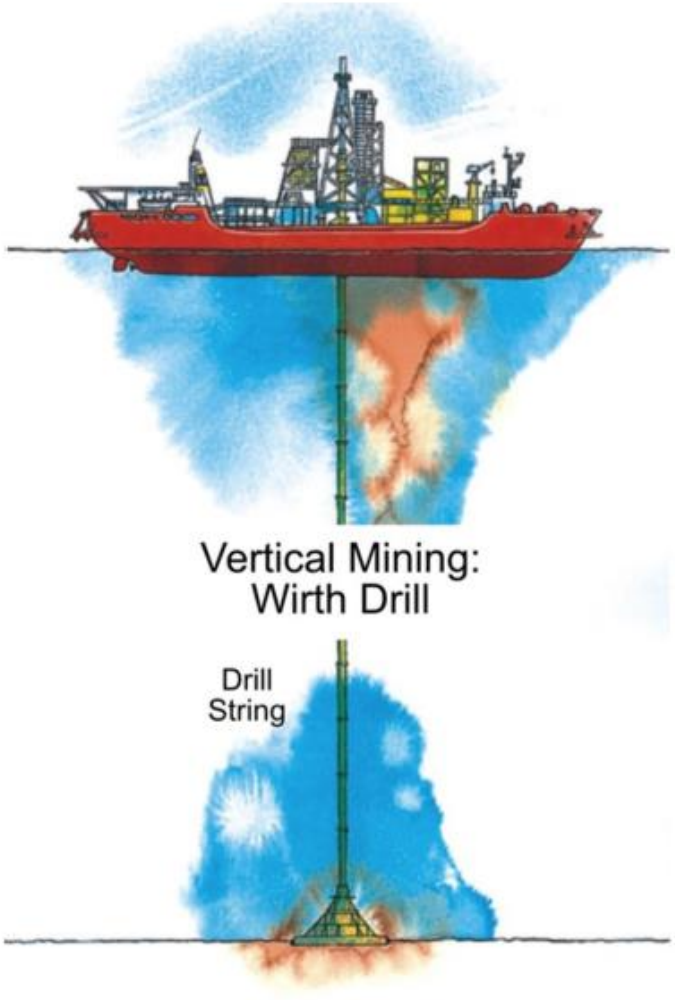
Medenbach 2000

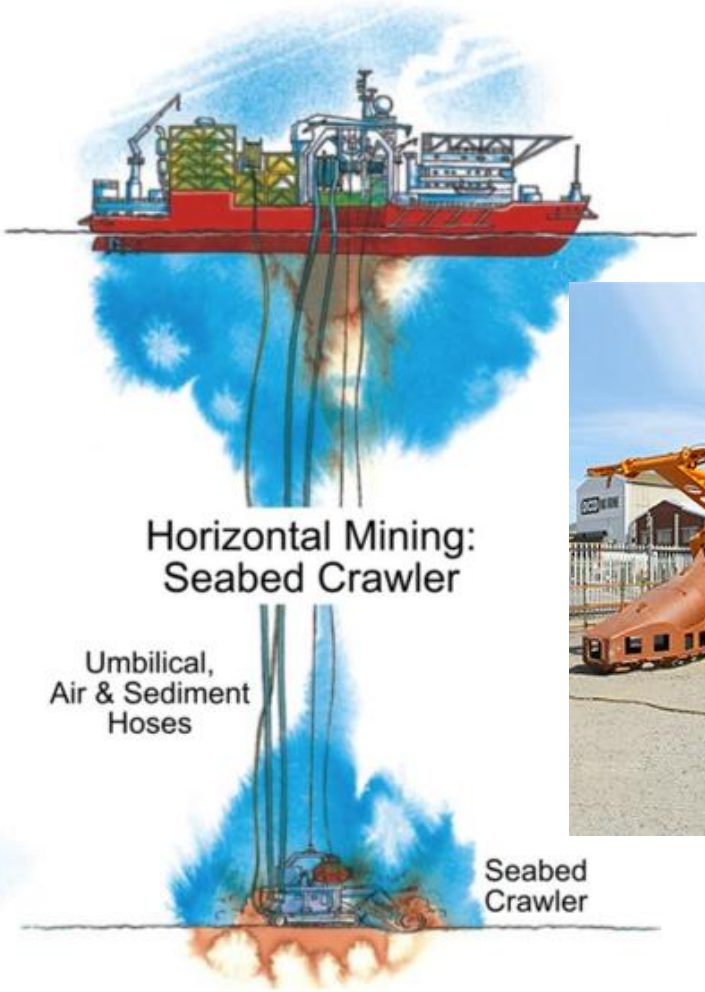




- Bedrock is literally ‘wiped clean’
- Operations below sea level behind sea walls

Production
Alluvial and marine diamond mines



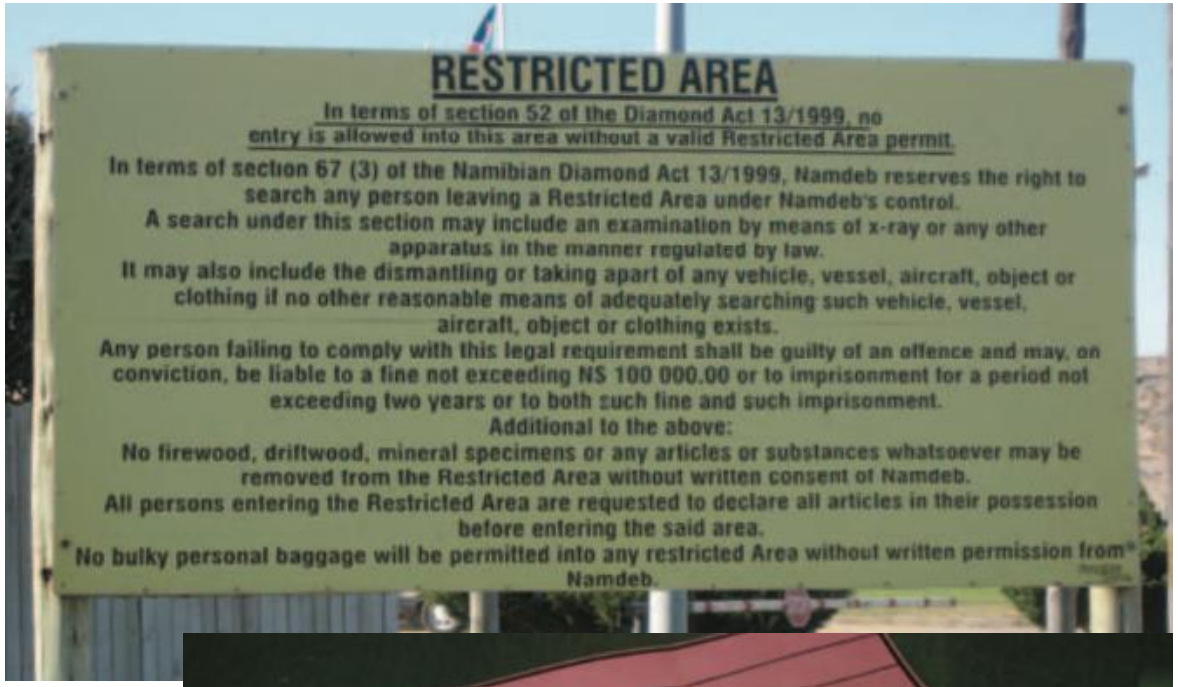




© de Beers Marine

- Drilling/sweeping or dredging up sediment from sea floor (<140m deep)
- Impact on sea life?

Production
Beneficiation





- Crushing
- Pans and cyclones to concentrate heavy minerals
- Fine material on grease tables
- Coarser material separated via X-ray sorters with air jets

Kampf 2007

Diamond deposits
Key learning points week 5

- Diamonds (pure C) are the hardest material we know, and are used mainly as abrasive and as gemstones.
- Diamonds form in the mantle below deep lithospheric keels of old cold crust – evidence: inclusions of mantle material, density of diamonds.
- Explosive volcanism brings diamonds to surface – kimberlite and lamproite pipes. These are windows into mantle conditions. Most are intracontinental volcanics.
- Primary diamond deposits occur in volcanic pipes, secondary deposits are placer.
- Links exist between kimberlite volcanism, plate tectonics and mantle plumes.

